10x 单细胞基因编辑测序实验经验分享

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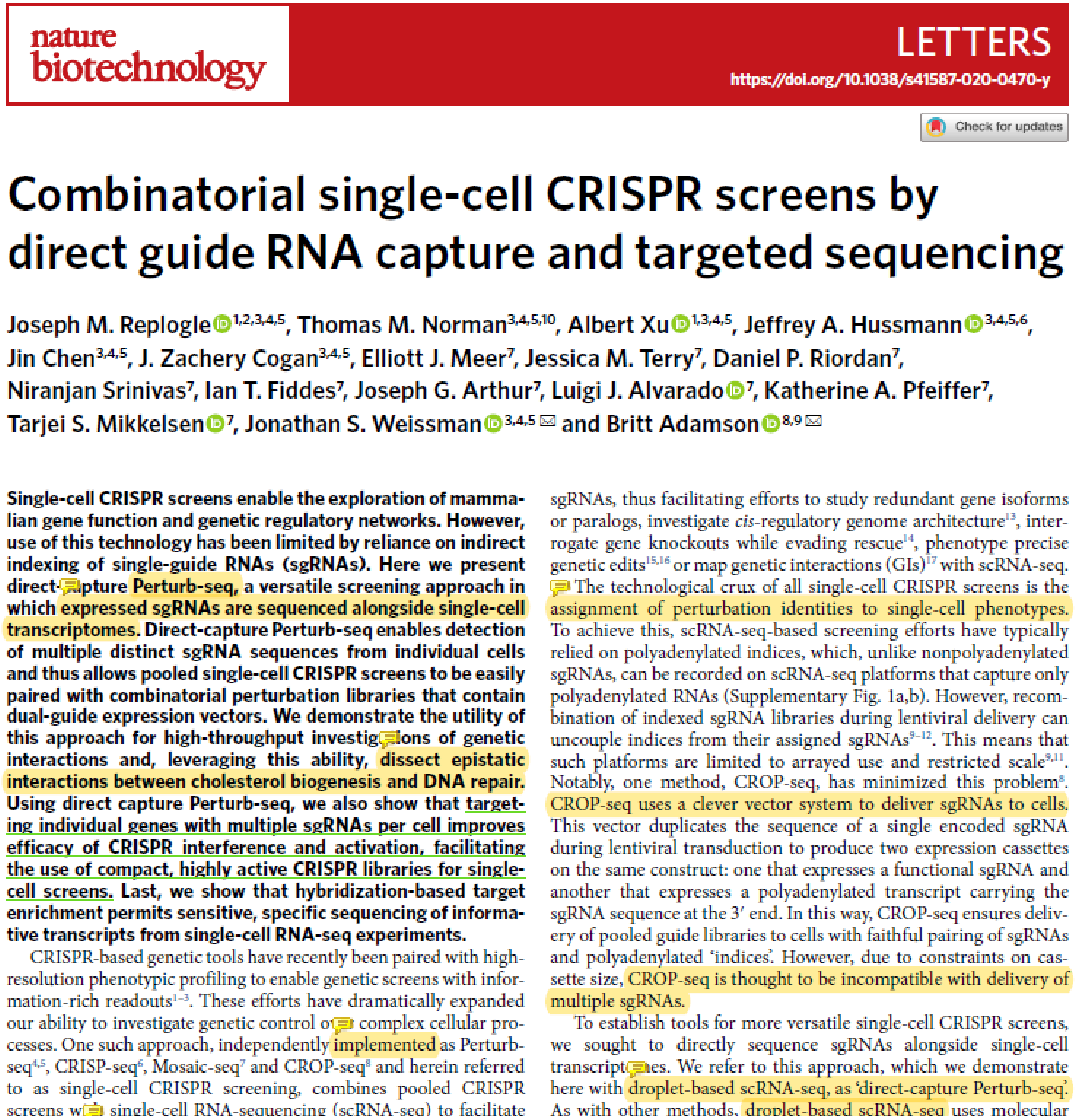

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LETTERS

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Check for updates

Combinatorial single-cell CRISPR screens by direct guide RNA capture and targeted sequencing

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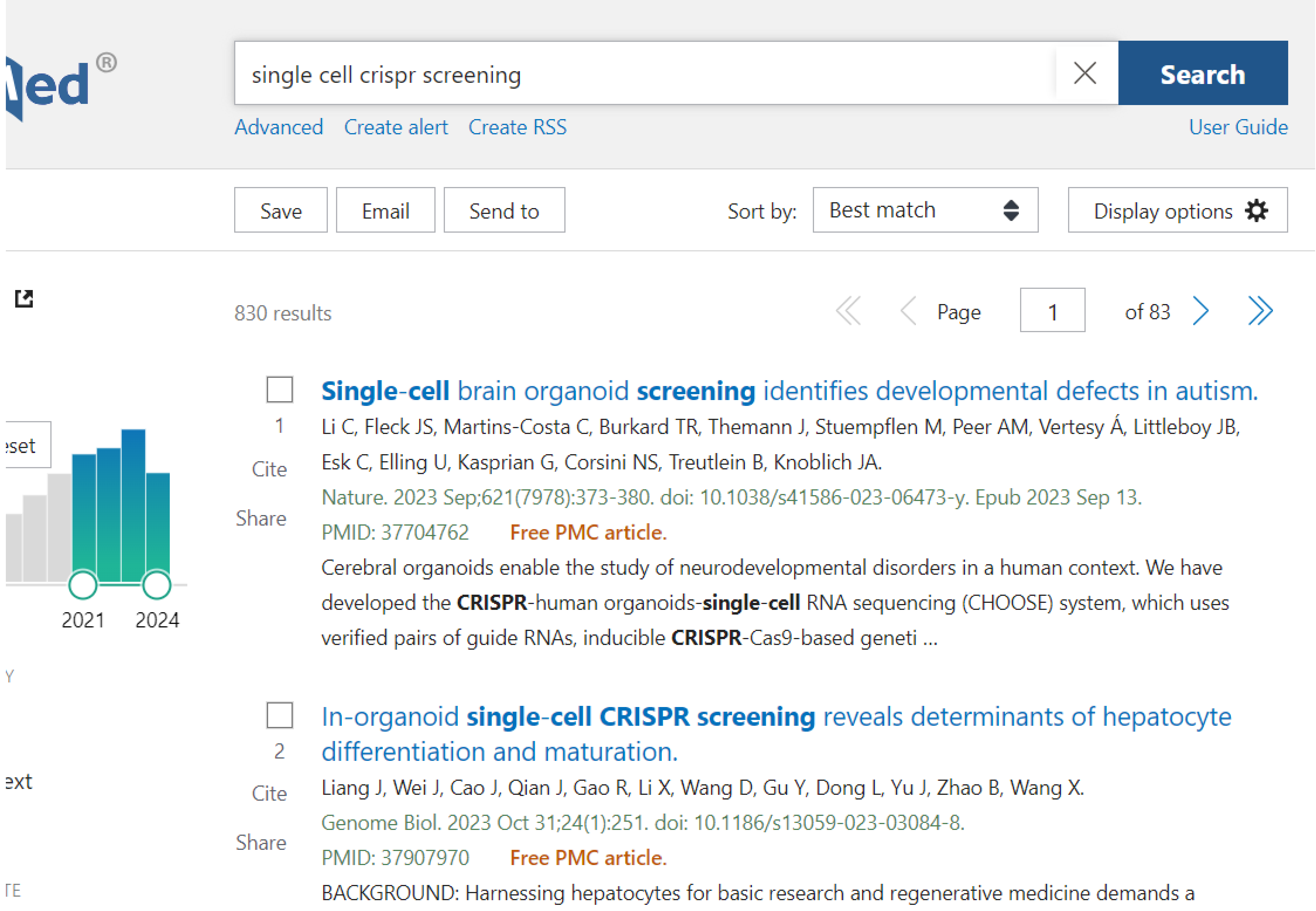
Single-cell CRISPR screens enable the exploration of mammalian gene function and genetic regulatory networks. However, use of this technology has been limited by reliance on indirect indexing of single-guide RNAs (sgRNAs). Here we present direct-capture Perturb-seq, a versatile screening approach in which expressed sgRNAs are sequenced alongside single-cell transcriptomes. Direct-capture Perturb-seq enables detection of multiple distinct sgRNA sequences from individual cells and thus allows pooled single-cell CRISPR screens to be easily paired with combinatorial perturbation libraries that contain dual-guide expression vectors. We demonstrate the utility of this approach for high-throughput investigations of genetic interactions and, leveraging this ability, dissect epistatic interactions between cholesterol biogenesis and DNA repair. Using direct capture Perturb-seq, we also show that targeting individual genes with multiple sgRNAs per cell improves efficacy of CRISPR interference and activation, facilitating the use of compact, highly active CRISPR libraries for single-cell screens. Last, we show that hybridization-based target enrichment permits sensitive, specific sequencing of informative transcripts from single-cell RNA-seq experiments.

CRISPR-based genetic tools have recently been paired with high-resolution phenotypic profiling to enable genetic screens with information-rich readouts[1–3]. These efforts have dramatically expanded our ability to investigate genetic control of complex cellular processes. One such approach, independently implemented as Perturb-seq[4,5], CRISP-seq[6], Mosaic-seq[7] and CROP-seq[8] and herein referred to as single-cell CRISPR screening, combines pooled CRISPR screens with single-cell RNA-sequencing (scRNA-seq) to facilitate sgRNAs, thus facilitating efforts to study redundant gene isoforms or paralogs, investigate *cis*-regulatory genome architecture[13], interrogate gene knockouts while evading rescue[14], phenotype precise genetic edits[15,16] or map genetic interactions (GIs)[17] with scRNA-seq. The technological crux of all single-cell CRISPR screens is the assignment of perturbation identities to single-cell phenotypes. To achieve this, scRNA-seq-based screening efforts have typically relied on polyadenylated indices, which, unlike nonpolyadenylated sgRNAs, can be recorded on scRNA-seq platforms that capture only polyadenylated RNAs (Supplementary Fig. 1a,b). However, recombination of indexed sgRNA libraries during lentiviral delivery can uncouple indices from their assigned sgRNAs[9–12]. This means that such platforms are limited to arrayed use and restricted scale[9,11]. Notably, one method, CROP-seq, has minimized this problem[8]. CROP-seq uses a clever vector system to deliver sgRNAs to cells. This vector duplicates the sequence of a single encoded sgRNA during lentiviral transduction to produce two expression cassettes on the same construct: one that expresses a functional sgRNA and another that expresses a polyadenylated transcript carrying the sgRNA sequence at the 3′ end. In this way, CROP-seq ensures delivery of pooled guide libraries to cells with faithful pairing of sgRNAs and polyadenylated 'indices'. However, due to constraints on cassette size, CROP-seq is thought to be incompatible with delivery of multiple sgRNAs.

To establish tools for more versatile single-cell CRISPR screens, we sought to directly sequence sgRNAs alongside single-cell transcriptomes. We refer to this approach, which we demonstrate here with droplet-based scRNA-seq, as 'direct-capture Perturb-seq'. As with other methods, droplet-based scRNA-seq uses molecular



目录

CRISPR Screen

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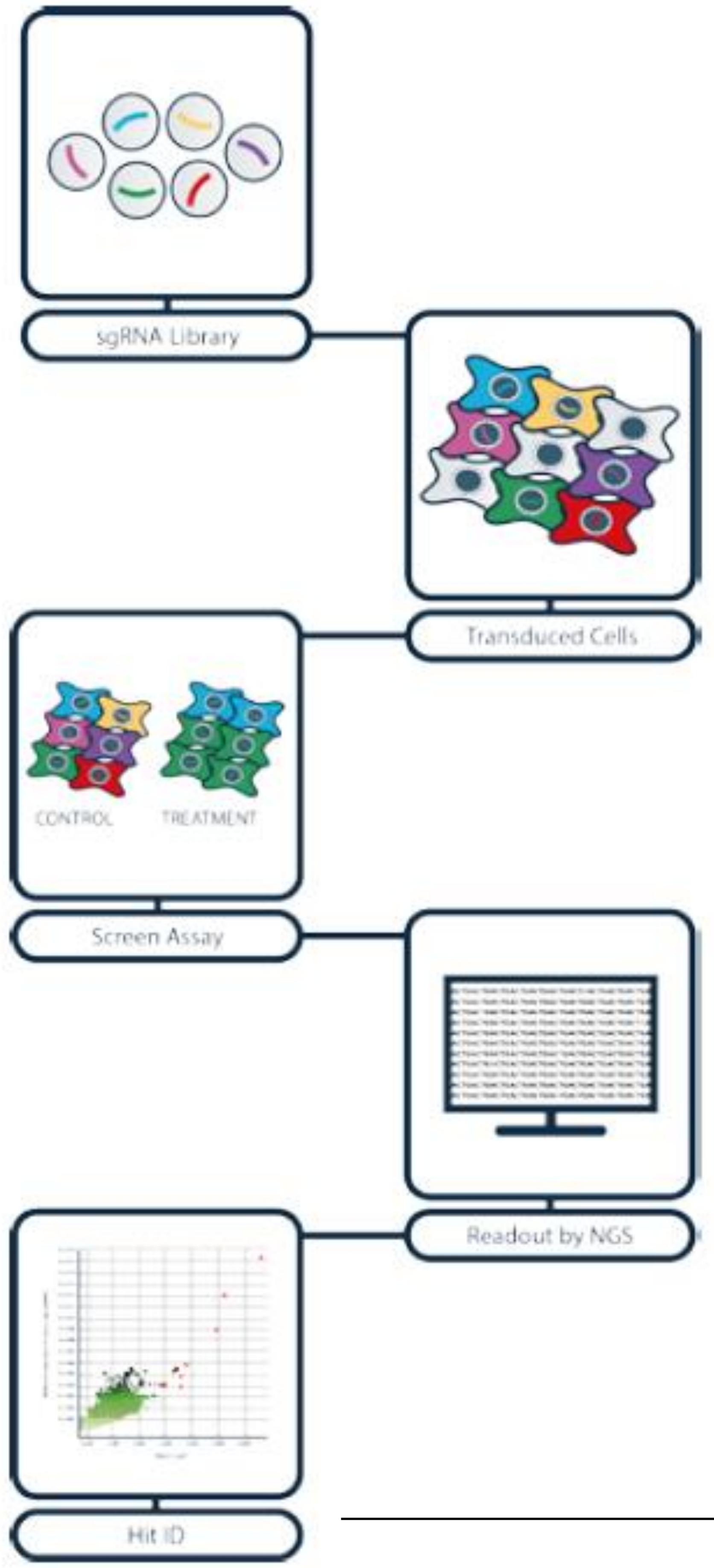
----大规模、高效进行“靶点研究” 的关键技术：CRISPR Screen



寻找新的药物靶点

HIV

CRISPR screen identifies novel therapeutic targets

sarah crunkhorn

Natuer Reviews Drug Discovery **16**,88 (2017) | doi:10.1038/nrd.2017.12

探寻药物作用机制

Cornerstones of CRISPR–Cas in drug discovery and therapy

Christof Fellmann, Benjamin G. Gowen, Pei-Chun Lin, Jennifer A. Doudna & Jacob E. Corn

Nature Reviews Drug Discovery **16**, 89–100 (2017)

doi:10.1038/nrd.2016.238

开发联合用药

Synergistic drug combinations for cancer identified in aCRISPR screen for pairwise genetic interactions

Kyuho Han,Edwin E Jeng,Gaelen T Hess,David W morgens,Amy Li &Michael C Bassik

Affiliations | Contributions | Corresponding author

Natru Biotechnology **35**,463-474 (2017) | doi:10.1038/nbt.3834

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研究耐药机制

Functional Genomic Landscape of Human Breast Cancer Drivers, Vulnerabilities, and Resistance

•Richard Marcotte[1,9,12], Azin Sayad[1,9], Kevin R. Brown[2], Felix Sanchez-Garcia[4], Jüri Reimand[2,10], Maliha Haider[1], Carl Virtanen[1], James E. Bradner[5,6], Gary D. Bader[2], Gordon B. Mills[7], Dana Pe' er[4],

http://doi.org/10.1016/j.clell.2015.11.062

普通CRISPR Screen的缺点

1. 是DNA扩增测序，只能进行sgRNA序列的分析，即样本中含有多少种sgRNA和每种sgRNA的相对丰度
2. 最常用的是敲除库，很少做激活库，几乎没有抑制库
3. 无法进行基因编辑后转录组水平分析
4. 即使细胞群进行文库编辑，后续进行普通转录组建库，只能捕获mRNA序列，分析的也是细胞群的平均转录水平变化，不包含任何sgRNA信息；
5. 使用sgRNA和mRNA的RT引物进行转录组建库，片段大小差异太大，也需要把大小核酸片段分开建库，否则小片段的sgRNA序列都会被纯化丢失；如果分开建库，也无法将sgRNA序列和转录组水平变化相对应，不清楚是哪些sgRNA发挥的作用

普通CRISPR Screen难以解决以下科学问题

1. 哪些基因调控了某一个cell marker的表达?
2. 哪些基因会调控某个代谢通路?
3. 什么基因是某个病毒入侵的受体?
4. 哪些基因突变导致肿瘤细胞耐药?
5. 哪些基因的表达量会调控细胞分化方向?
6. 什么通路介导肿瘤细胞对化疗的敏感性?

这些都涉及到转录水平的解析

单细胞CRISPR（scCRISPR）是什么？

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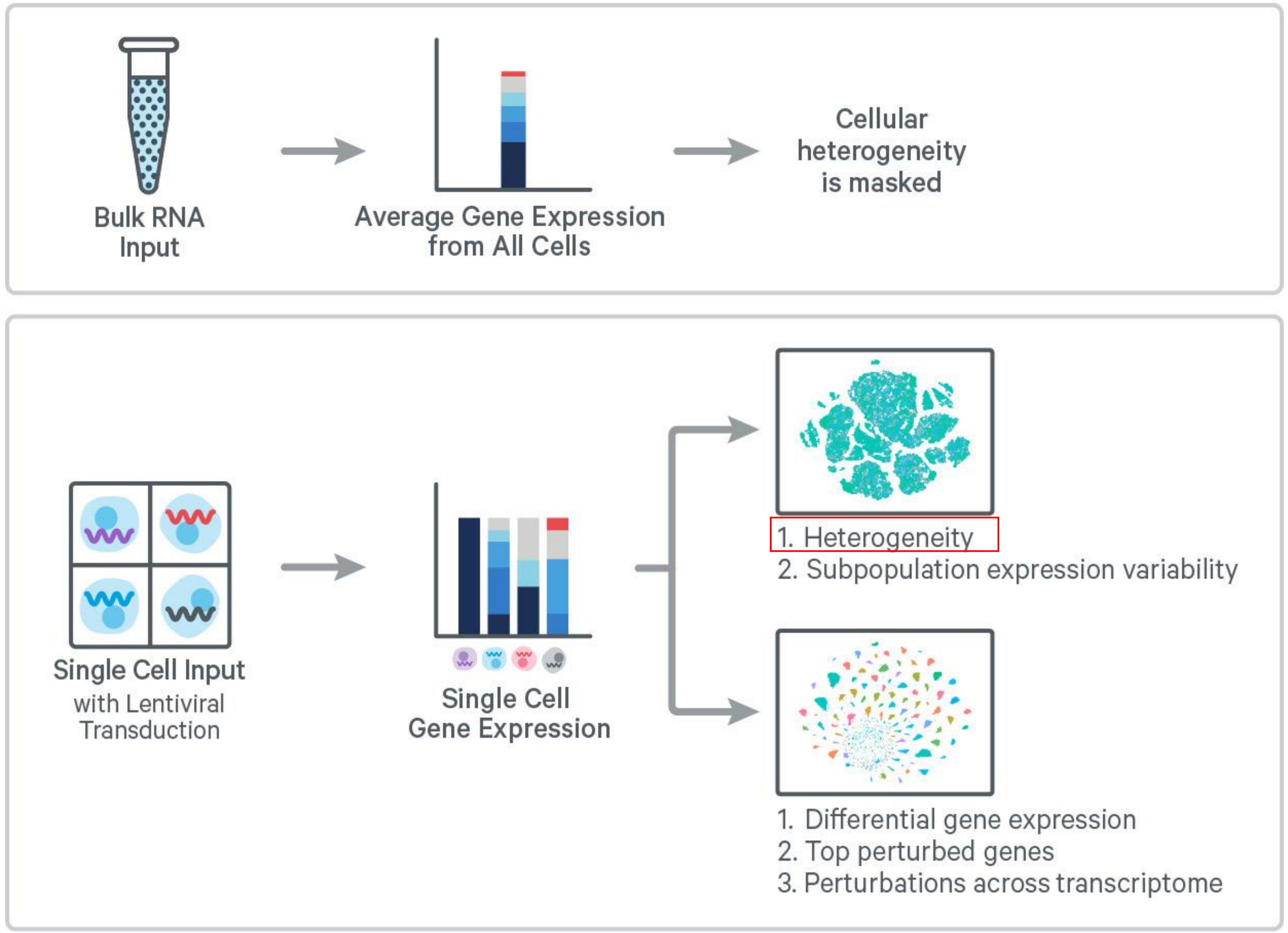
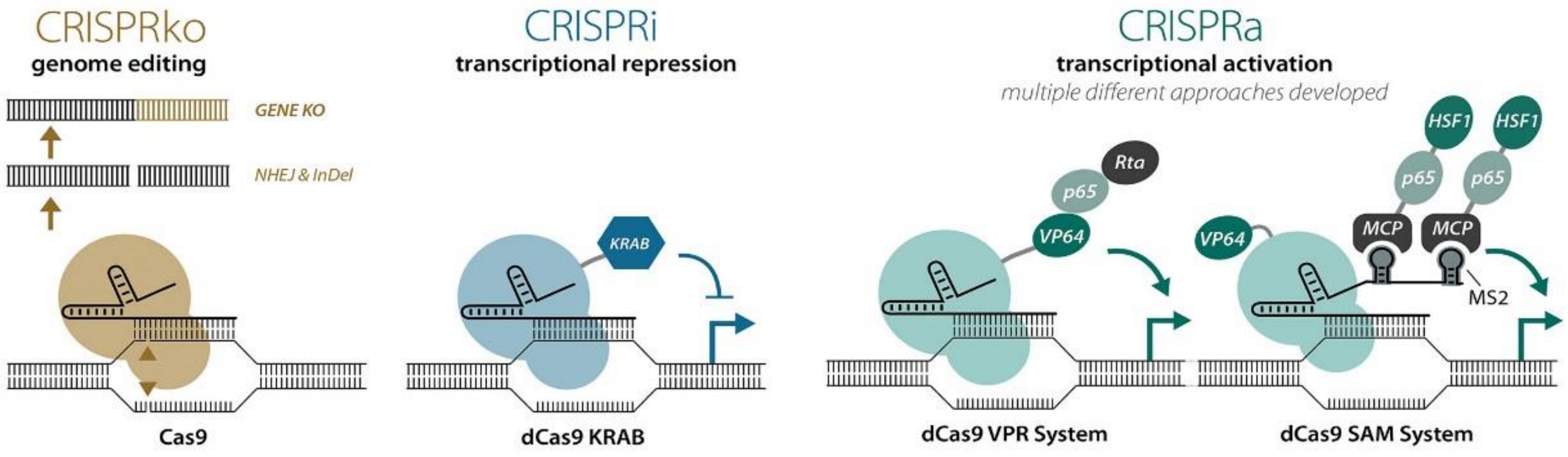
单细胞CRISPR文库筛选：将单细胞转录组测序与CRISPR文库筛选结合起来的一项技术

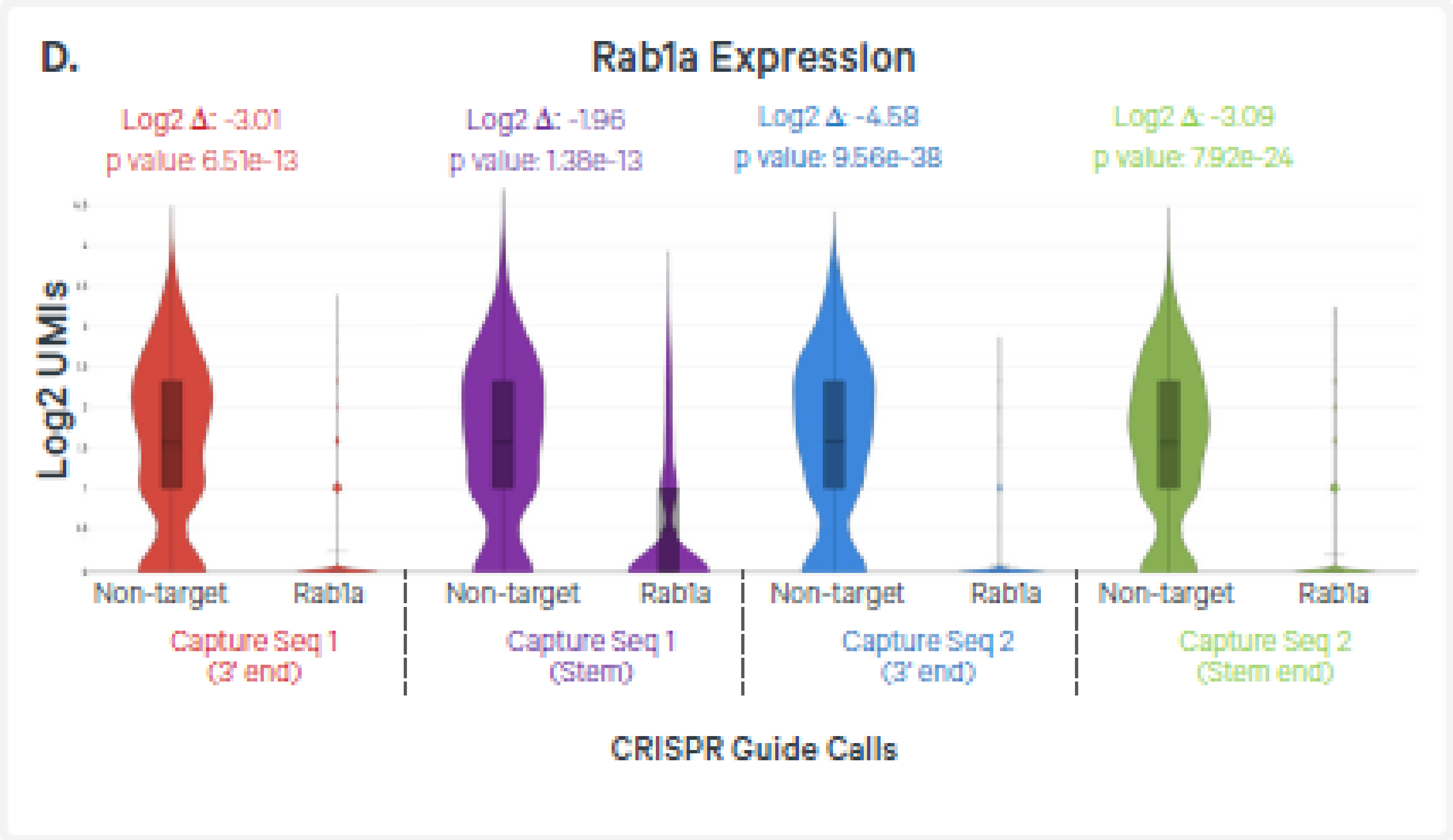
与普通单细胞CRISPR有什么不同？

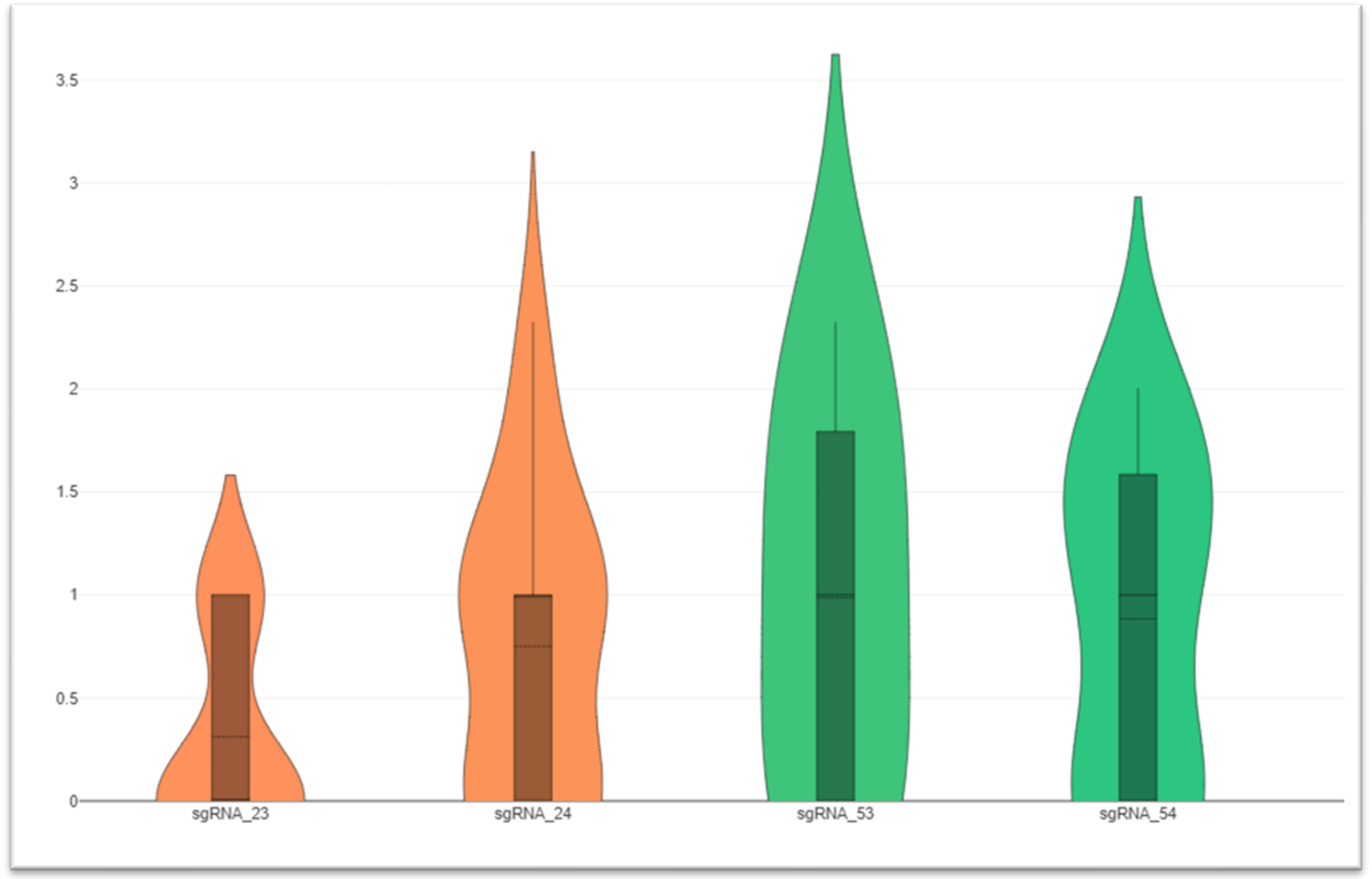


Chromium Single Cell Gene Expression coupled with Feature Barcode technology for CRISPR Screening provides a high-throughput and scalable approach to obtain gene expression profiles along with CRISPR-mediated perturbation phenotypes in the same single cell.

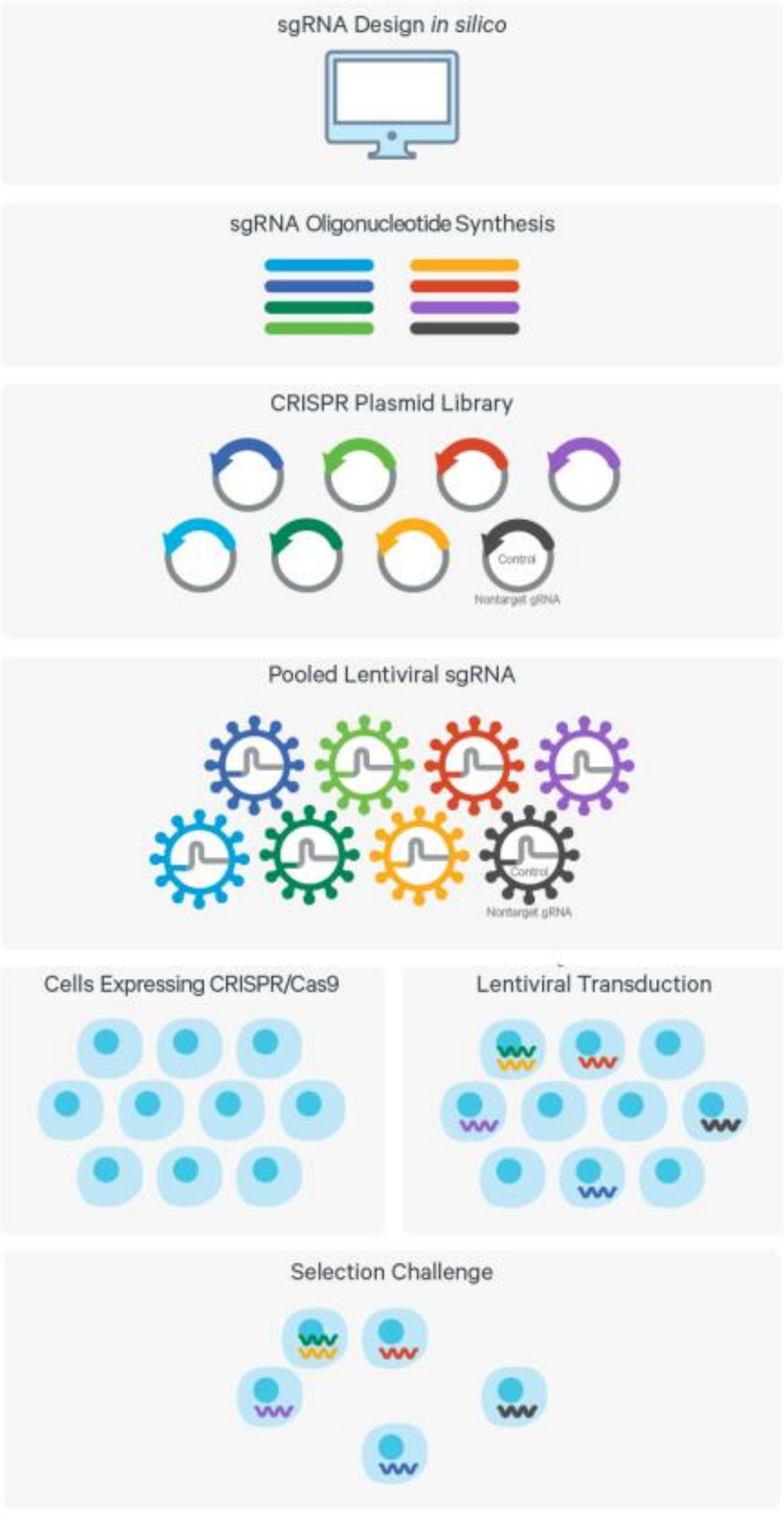
敲除库、抑制库和激活库的选择







scCRISPR细胞样本制备实验流程



设计sgRNA序列

合成sgRNA

构建sgRNA质粒文库

包装sgRNA慢病毒文库

sgRNA慢病毒文库感染稳转Cas9的细胞

筛选sgRNA-positive的细胞

NC sgRNA的使用比例及sgRNA文库大小的设计

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Control Non-targeted sgRNA per Gene

Typically 2-5 control non-targeting sgRNA can be used per experiment. For the 10x Genomics CRISPR Screening assay, the exact number of control sgRNAs used isn't important. 10x Genomics recommends a targeted recovery of 500-1,000 cells that include non-target sgRNAs. For example, if using a single Chromium Chip channel for the assay with a targeted recovery of 10,000 cells, 5-10% of the cells should contain non-target sgRNA(s).

推荐数目

- 每个Target gene 2-5条sgRNA。
- Non-target sgRNA：每个实验2-5条（但是对于10x Genomics CRISPR Screening只考虑含有Non-target sgRNA的捕获细胞数目（500-1000个）即可，NC sgRNA种类不重要）。

Assay Cell Load

10x Genomics recommends using 100-200 cells per targeting sgRNA. This ensures enough statistical power to be able to determine the significance of the perturbation. For non-targeting sgRNAs, which are critical for providing a baseline for calculating perturbations, using 500-1,000 cells is recommended.

Chromium Chip

Cells per channel in the chip
The Chromium Single Cell 3' Gene Expression with CRISPR Screening currently supports recovering 500-10,000 cells per channel of Chromium Chip used for the protocol.

Perturbations measured using a single channel
If a user loads the maximum number of cells (10,000) in a single channel of a chip, 500-1,000 (5-10%) of these cells should be non-targeting sgRNA containing cells.

Each targeting sgRNA should be represented by 100-200 cells (1-2%). Using this setup, ~45-90 sgRNA can be tested using a single channel.

Expanding the number of perturbation/cells beyond recommendations by 10x Genomics
Using the same recommendation as the preceding section, 500-1,000 total cells should be made up of non-targeting containing guides and each perturbation be represented by 100-200 cells. If all 8 channels in a chip are used with recovery of ~80,000 cells, this would enable a user to generate a pooled CRISPR screen so that ~1,000 cells (1.25%) contain non-targeting sgRNA and each targeting sgRNA would make up ~0.2% of the pool, allowing for testing of ~500 sgRNA.

单细胞实验流程

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材料选取 → 单细胞悬液制备并收集GEMs → 逆转录获得cDNA → 文库构建 → Nova 2*150测序 → 数据分析

送样要求

01 **细胞数量**

建议50万，最低尝试10万

04 **细胞浓度**

700~1200cell/μL

02 **细胞活性**

大于85%

05 **细胞状态**

无明显细胞团块和杂质；细胞状态稳定

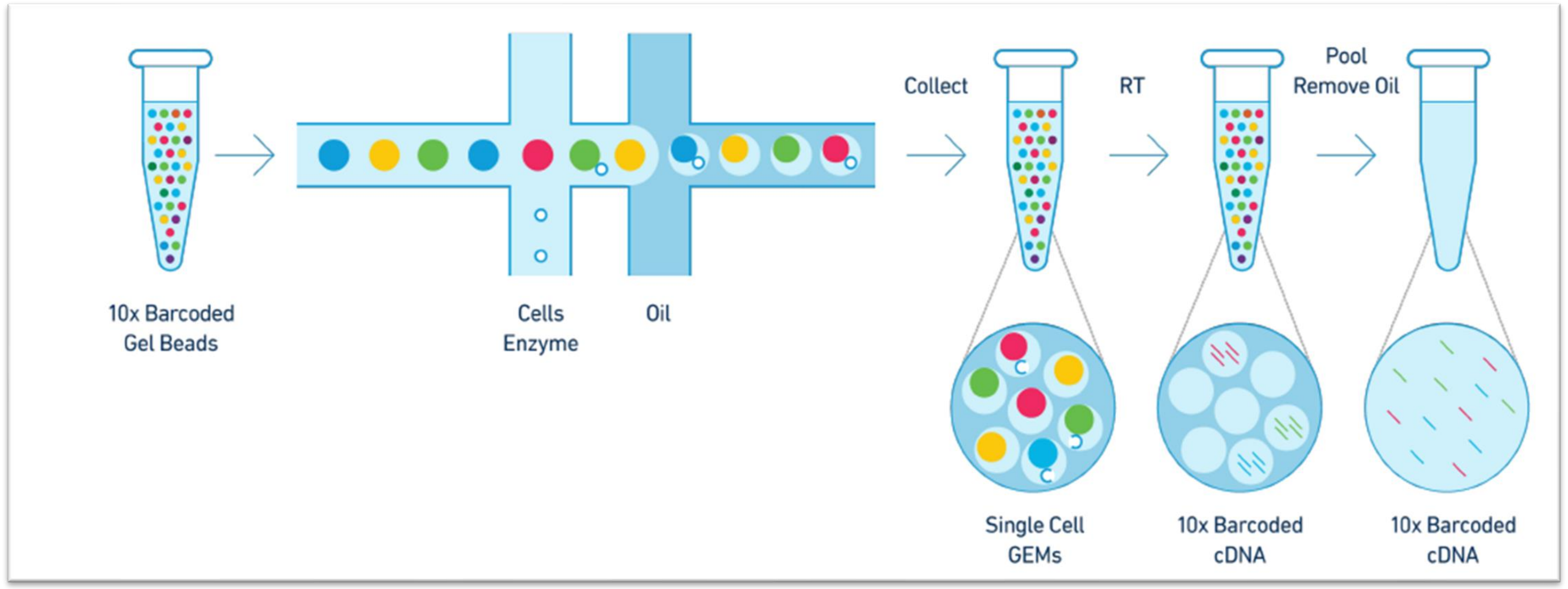
03 **细胞直径**

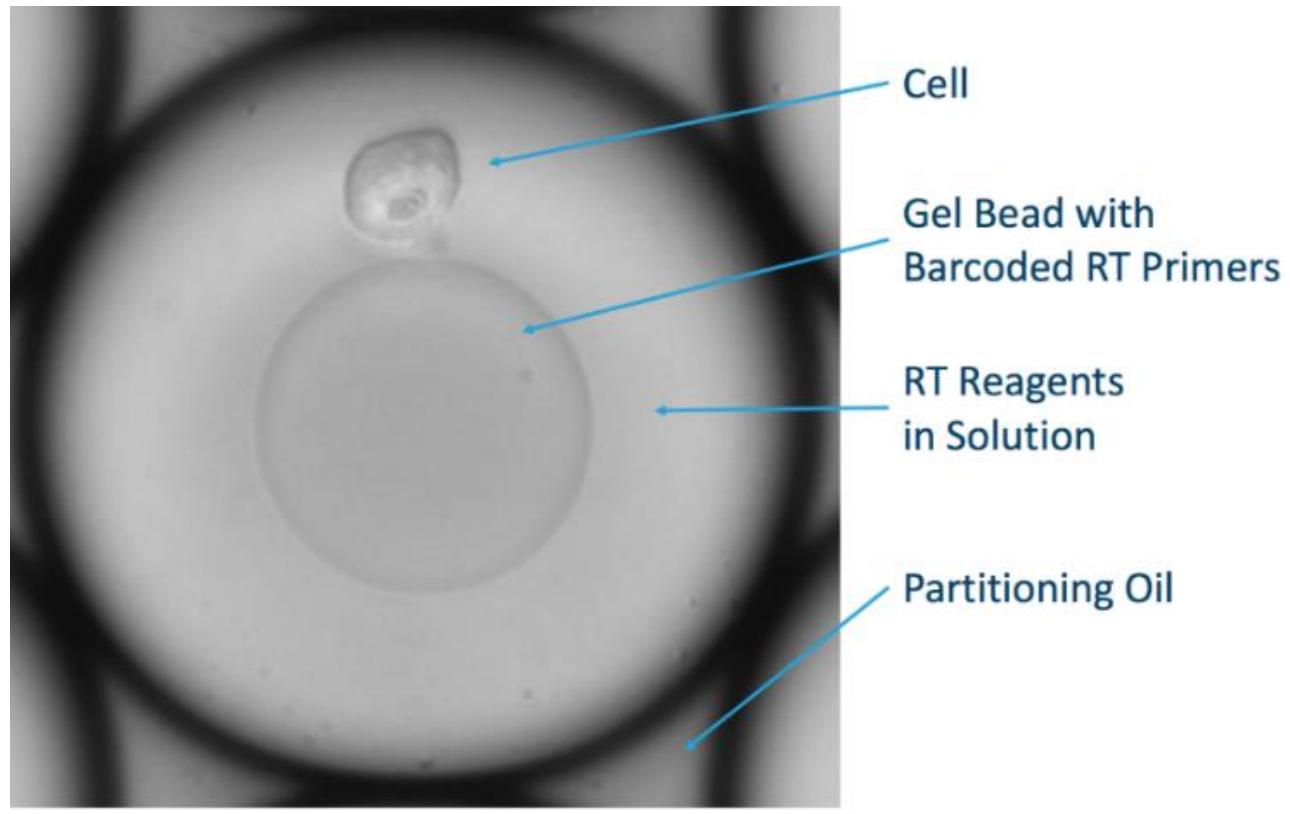
5-40μM或细胞核

06 **缓冲液**

缓冲液中不能含Ca2+和Mg2+等酶抑制剂

10X Genomics Chromium原理





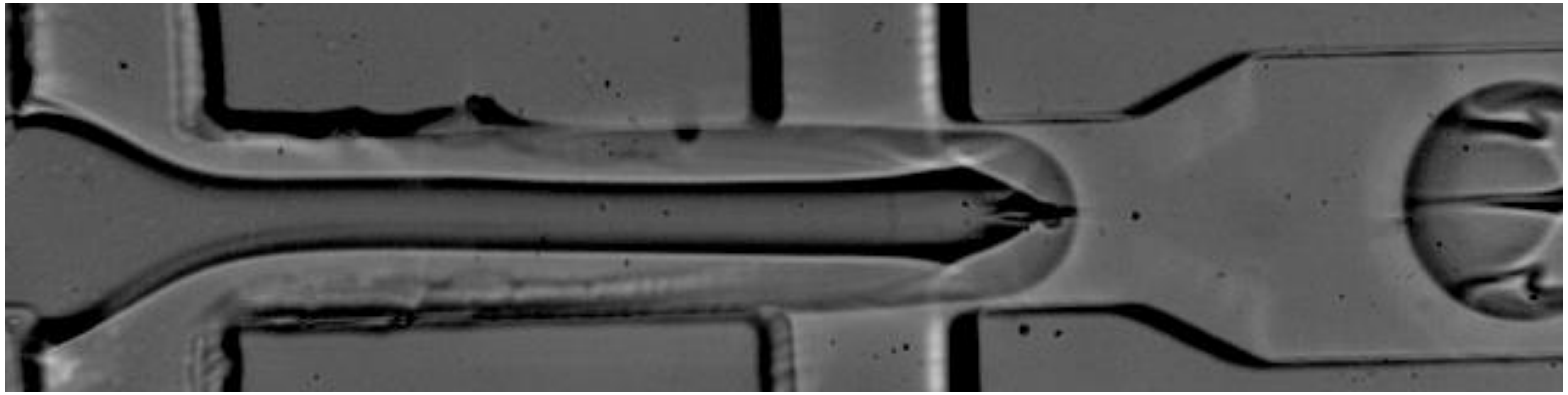


Image adapted from: Evan Z. Macosko, et al. Highly Parallel Genome-wide Expression Profiling of Individual Cells Using Nanoliter Droplets, Cell, Volume 161, Issue 5, 2015, Pages 1202-1214, ISSN 0092-8674, https://doi.org/10.1016/j.cell.2015.05.002. (http://www.sciencedirect.com/science/article/pii/S0092867415005498)

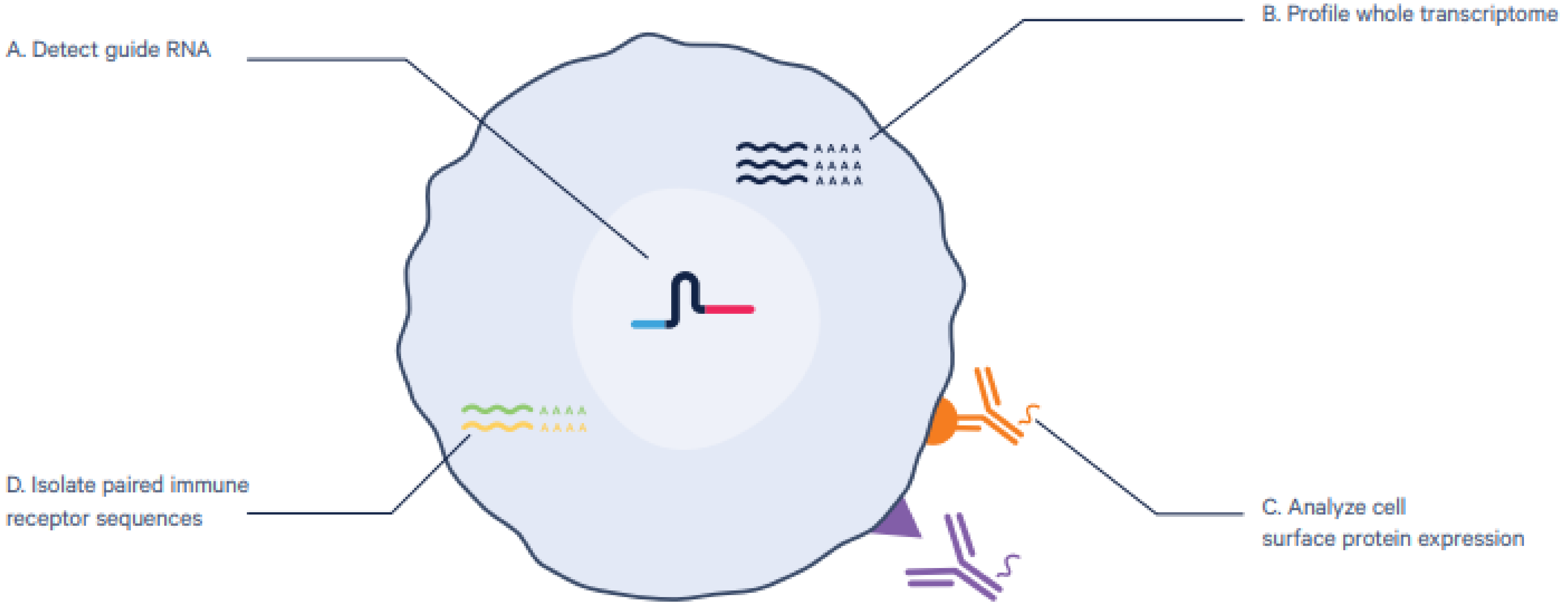
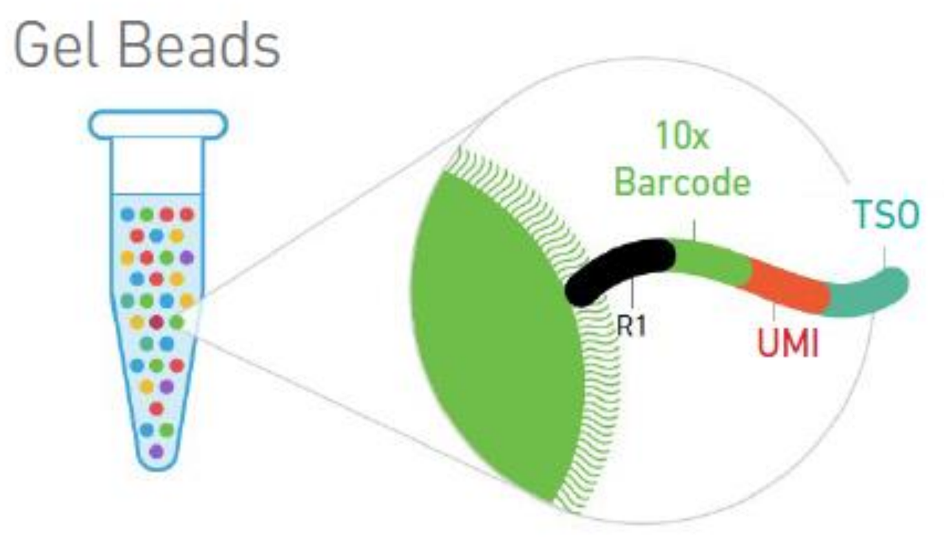


Figure 1. Chromium Single Cell CRISPR Screening enables high-throughput perturbation studies. Characterize perturbations at an even deeper level with Single Cell 5' Gene Expression, enabling detection of up to four modalities at single cell resolution, including guide RNAs (**A**), whole transcriptome gene expression (**B**), cell surface protein expression (**C**), and immune receptor sequences (**D**).

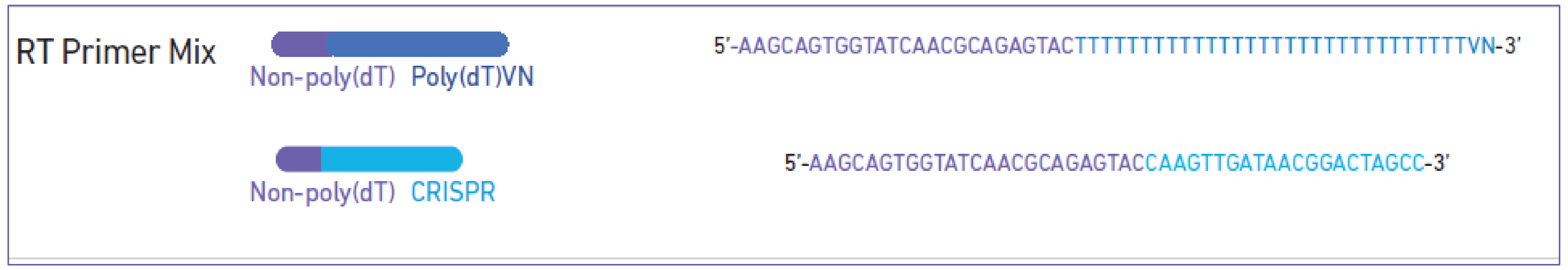
Highlights

- Evaluate entire gene networks, key disease pathways, or specific gene functions with multiomic, single cell profiling of up to thousands of CRISPR-driven gene edits
- Perform large-scale pathway screens to access more information per target, enabling more confident decision making for drug candidates
- Validate hits from genome-wide association studies (GWAS) and obtain comprehensive functional information from CRISPR edits to elucidate gene function

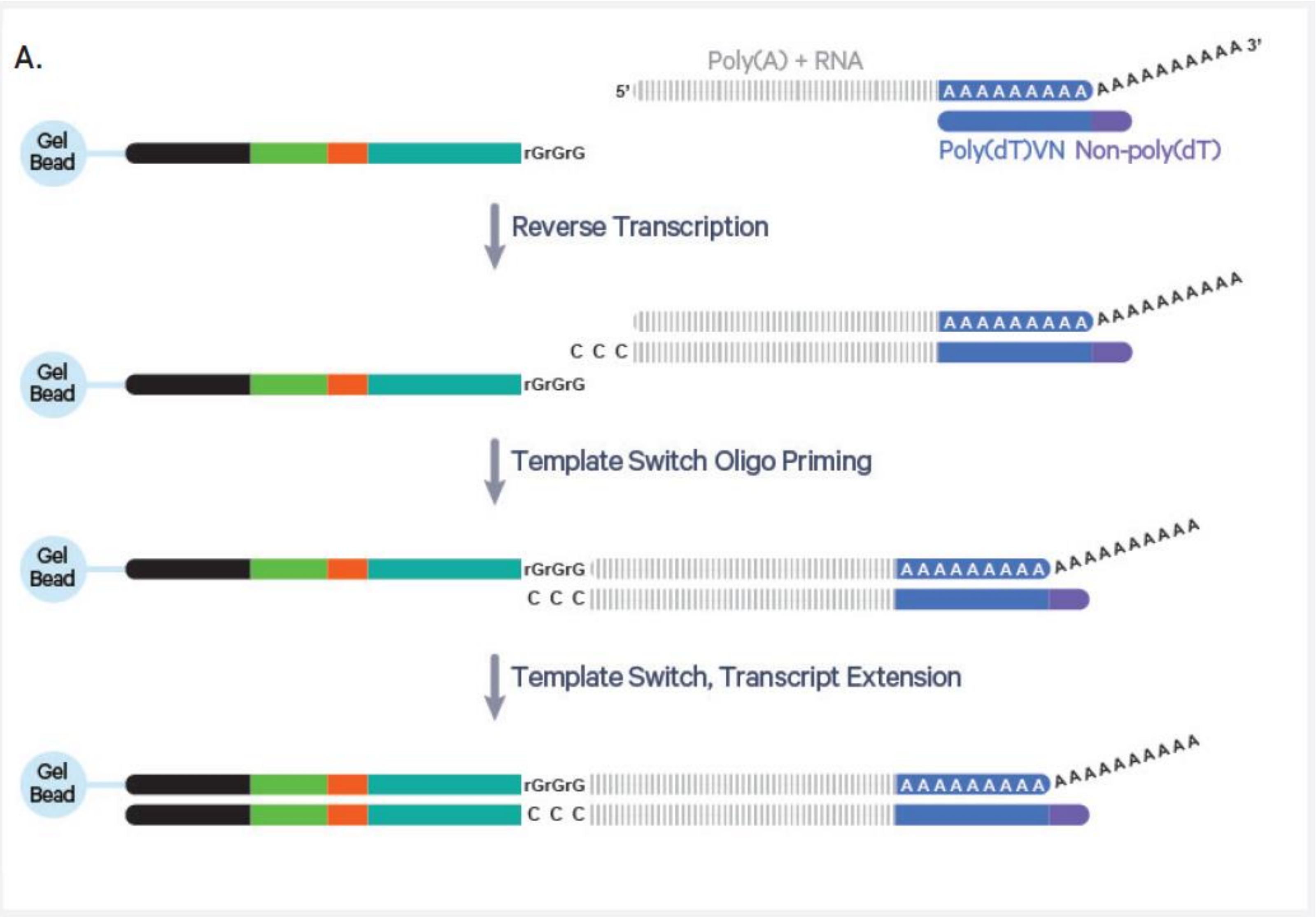
5’ Single Cell CRISPR Screen RT原理

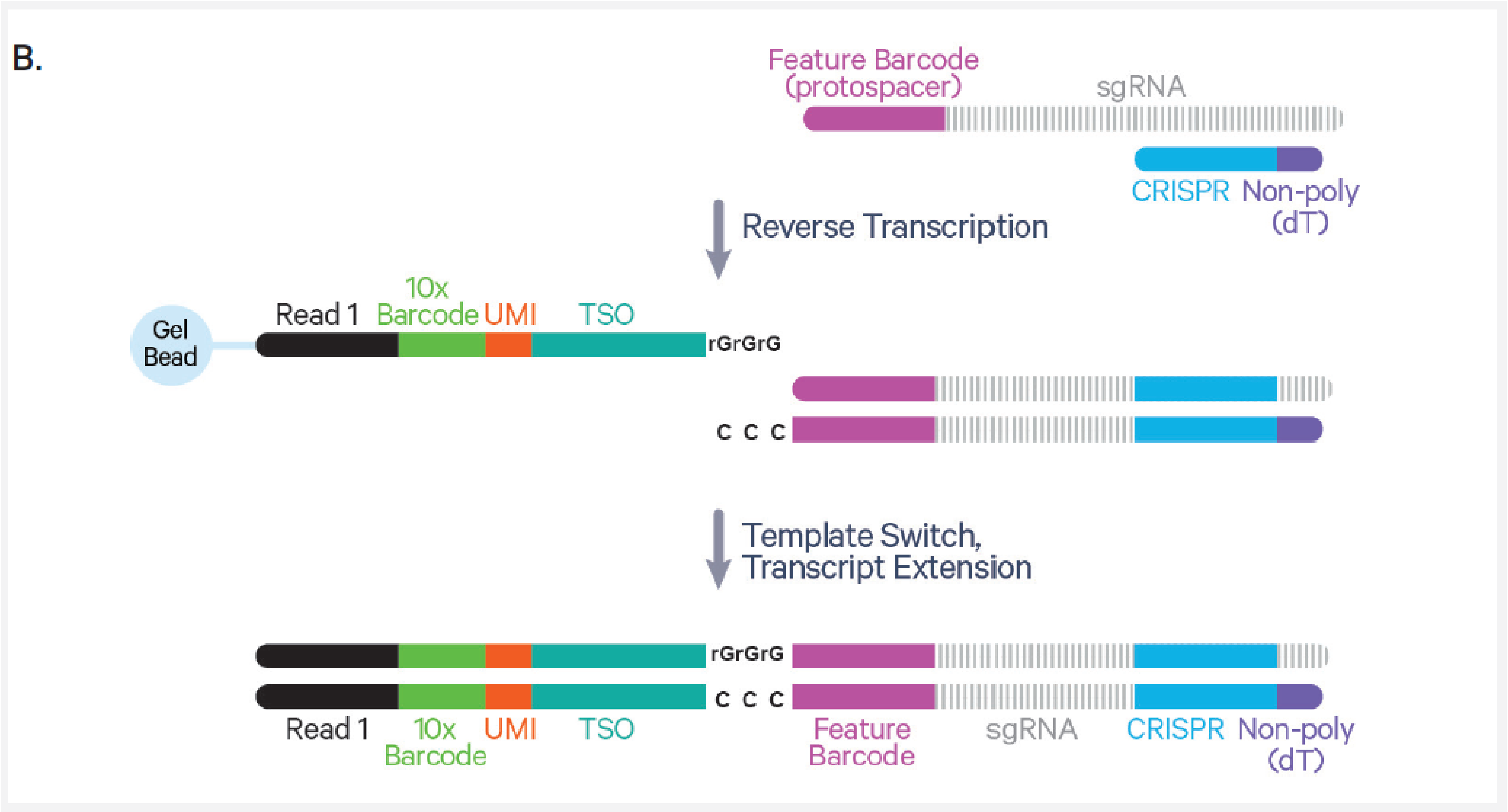


sgRNA结构示意图



Inside individual GEMs



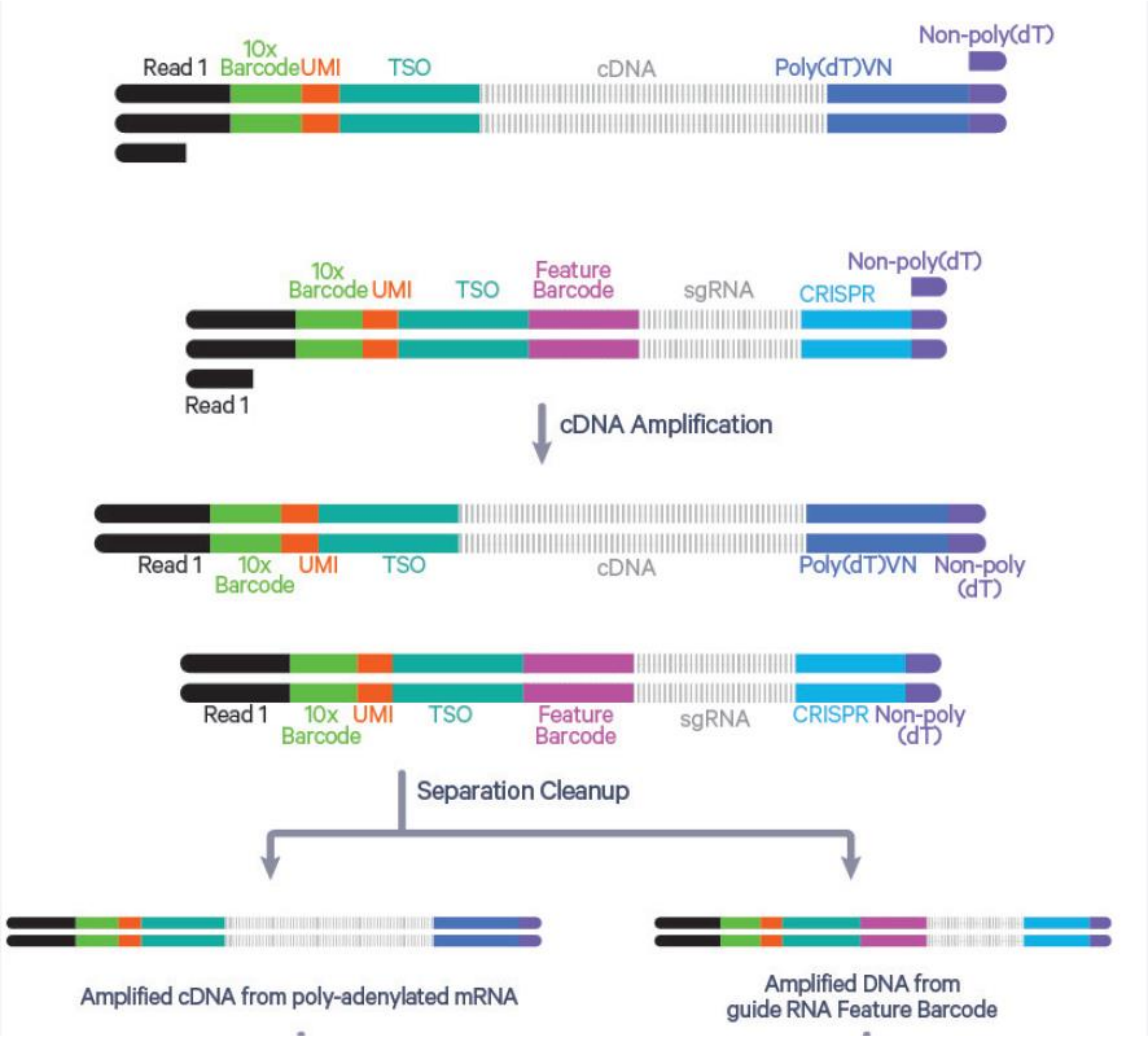


5’ ScCRISPR gRNA骨架序列注意事项

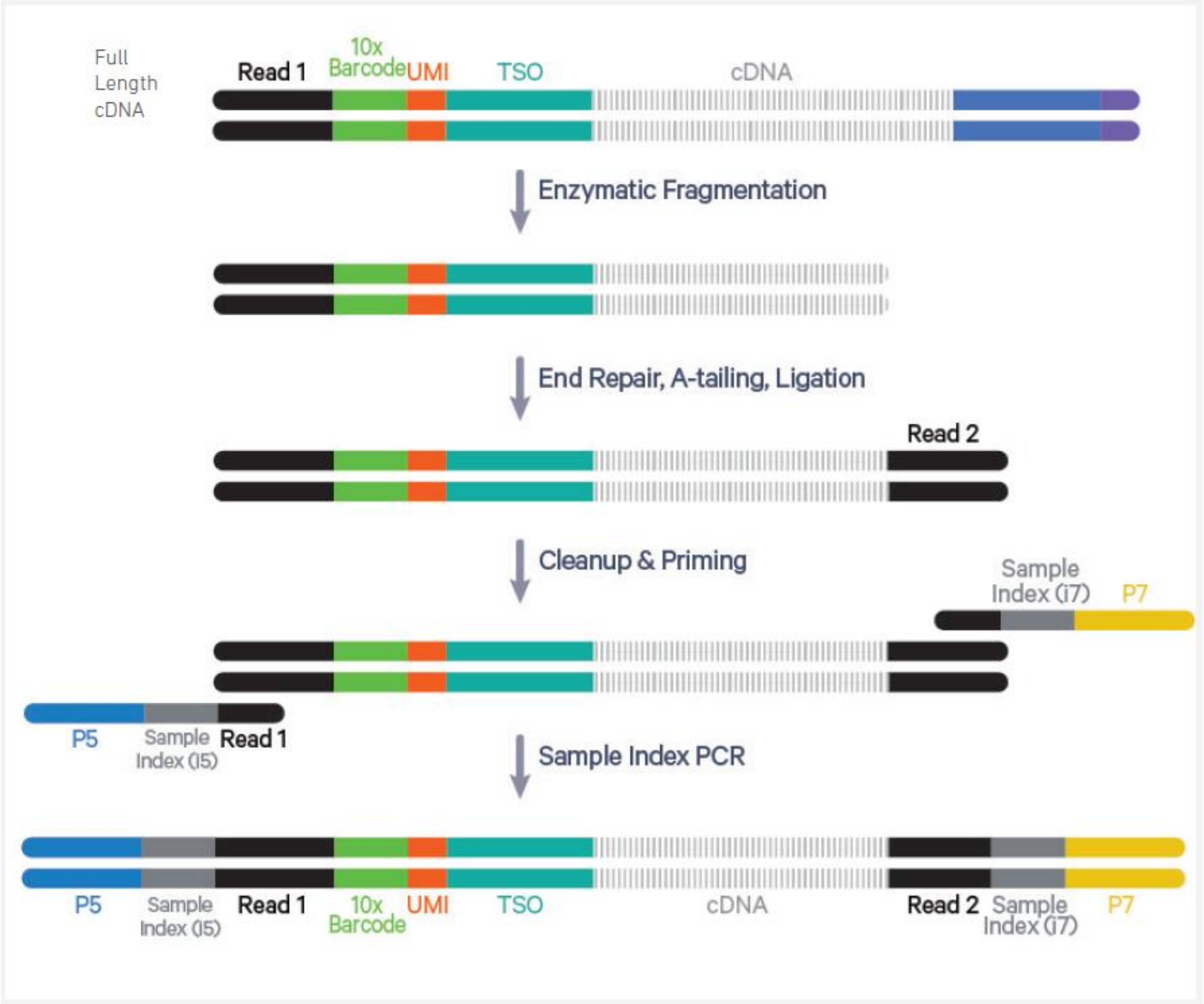
转录方向上面一条链捕获片段末端需要包含aataaggctagtccgttatcaacttg

5’ Single Cell CRISPR Screen建库原理

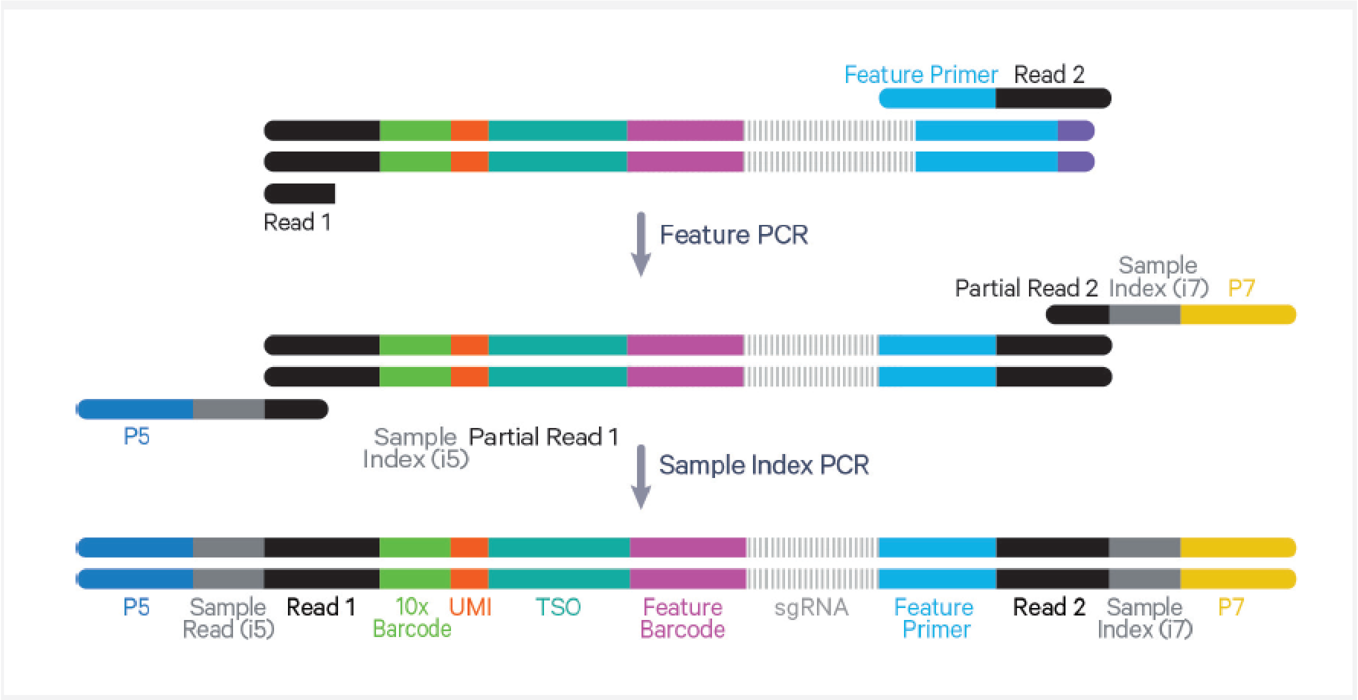
Pooled cDNA amplification



5' Gene Expression (GEX) Library Construction



5' CRISPR Screening Library Construction



3’ Single Cell CRISPR Screen建库原理

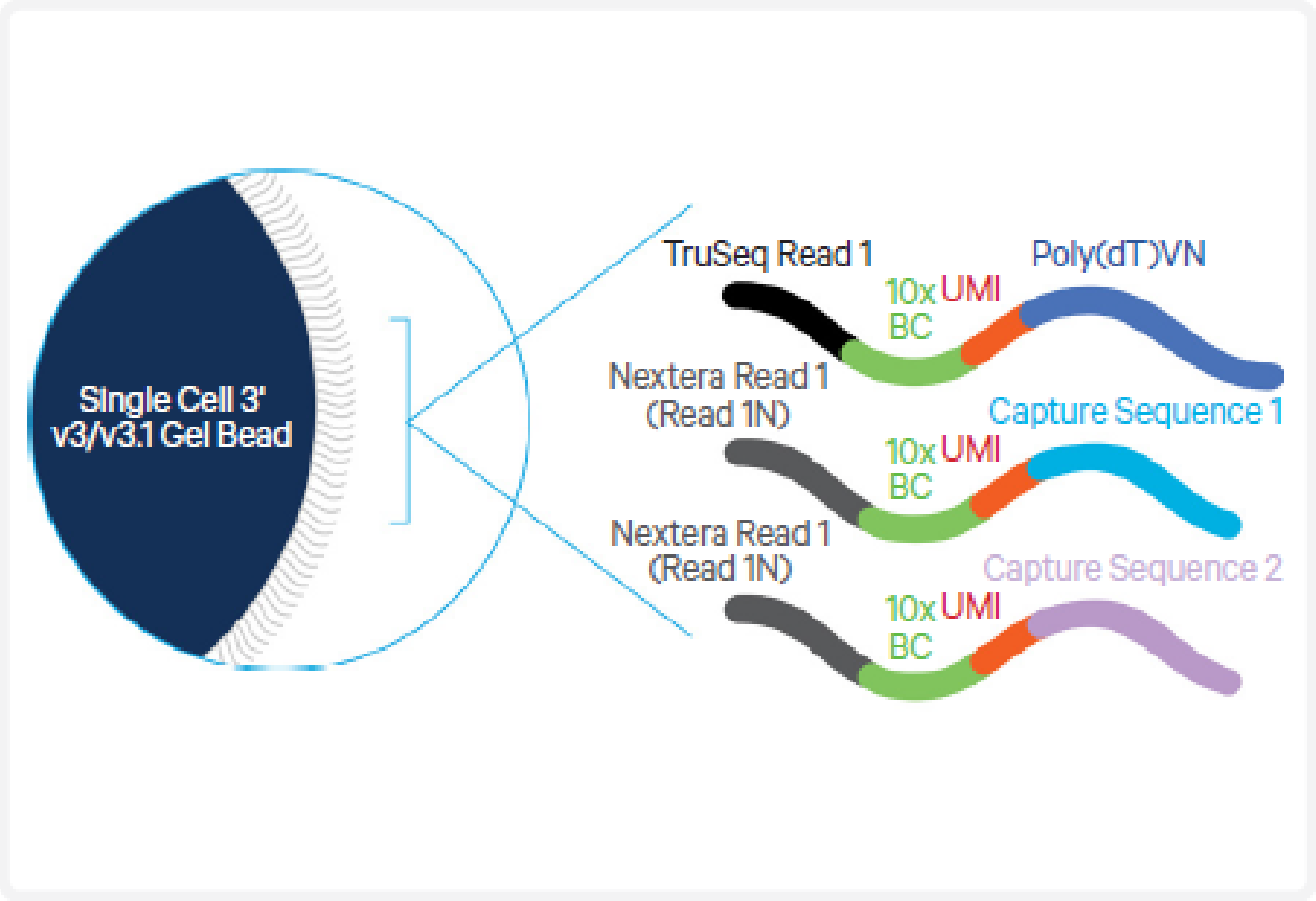
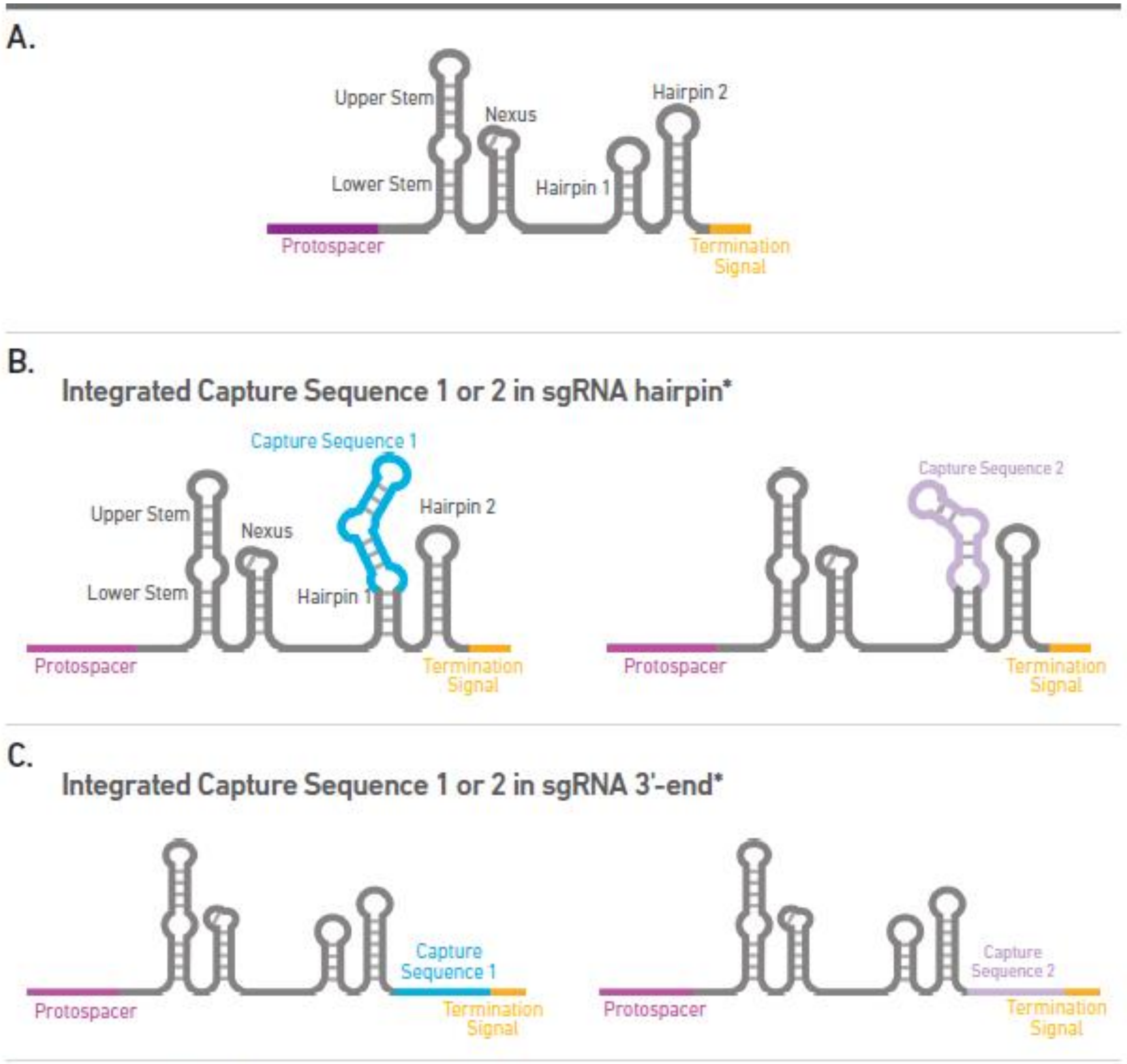


Figure 8. Chromium Single Cell 3’ v3/v3.1 Gel Bead Schematic.
Capture Sequence 1 and Capture Sequence 2 enable direct capture of sgRNA molecules from a cell inside a GEM.

Performing sgRNA QC by qPCR, NGS, or other methods is recommended prior to proceeding with the Single Cell Immune Profiling and CRISPR Screening Solution.



*Also compatible with Chromium Single Cell 3' CRISPR Screening assay.

5’ 与3’ 建库方式的选择

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How to decide if the 5’ CRISPR assay or the 3’ CRISPR assay best fit my experiment?

Question: How to decide if the 5’ CRISPR assay or the 3’ CRISPR assay best fit my experiment?

Answer: Sensitivity is comparable between the 3’ and 5’ CRISPR screening assays, both in terms of gene expression and guide capture.

The Single Cell 3' CRISPR Screening is enabled by engineering sgRNAs with one of two possible capture sequences that are required for assay compatibility. The Single Cell 5' CRISPR Screening approach is compatible with most Cas9 CRISPR vectors and does not require integration of specific capture sequences into the sgRNAs for compatibility.

In the majority of cases, customers should consider the 5’ CRISPR assay. The following criteria can help you decide which assay best fits your experiment.

Customers should select the 5’ CRISPR assay if they meet the following criteria:

- They already have a pre-built, compatible guide RNA library they want to use in the single cell CRISPR screening workflow.
- They are new to single cell CRISPR screening.
- They are experienced with single cell CRISPR screening, but do not wish to perform additional steps to modify guide libraries with assay-specific sequences.

Customers should select the 3’ CRISPR assay if they meet the following criteria:

- They want to compare their single cell CRISPR experiment to data from previous experiments that also used the 3’ assay. The same CRISPR assay should be used in order to minimize batch effects.
- They perform pilot experiments to determine the optimal choice of capture sequence and integration site for their cell type, as outlined in this article.

➢ 3’ 和5’ 的灵敏度、捕获效率是相当的

➢ 3’ 检测需要改造载体

➢ 大多数情况，推荐5’ 建库方式

◆普通单细胞转录组测序：

✓ 3’ 端的优势就是可以检测任何具有poly结构的RNA，例如mRNA，部分具有polyA结构的lnc RNA

✓ 5’ 端的优势就是可以获得免疫组库的信息，例如TCR和BCR的信息，因为这些基因重排和碱基突变都是靠近5端的

测序及分析

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数据量建议：

- ➢ mRNA文库-2W Reads per cell
- ➢ sgRNA文库-5K Reads per cell
- ➢ 需捕获细胞数：100-200 cell per sgRNA

以上为10×官方推荐最小测序量，需要根据具体实验调整。

我司测序数据量建议：

- ➢ mRNA文库：CC仪器：100G/通道；X仪器：200G/通道
- ➢ sgRNA文库：10G/通道； X仪器：20G/通道

需要根据具体需求调整。

CC与X平台捕获细胞数：

- ➢ CC仪器：
  - 单细胞转录组测序：1W cell/line左右
  - scCRISPR—5K-1.1W cell/line。
- ➢ X仪器：
  - 单细胞转录组测序：2W/line左右
  - scCRISPR—1.1W-2.5W cell/line

分析内容

原始数据 → 数据质控 → 单细胞层面的RNA丰度分析 → 细胞聚类分析 → 基于细胞聚类的sgRNA丰度分析 → gRNA-based细胞过滤

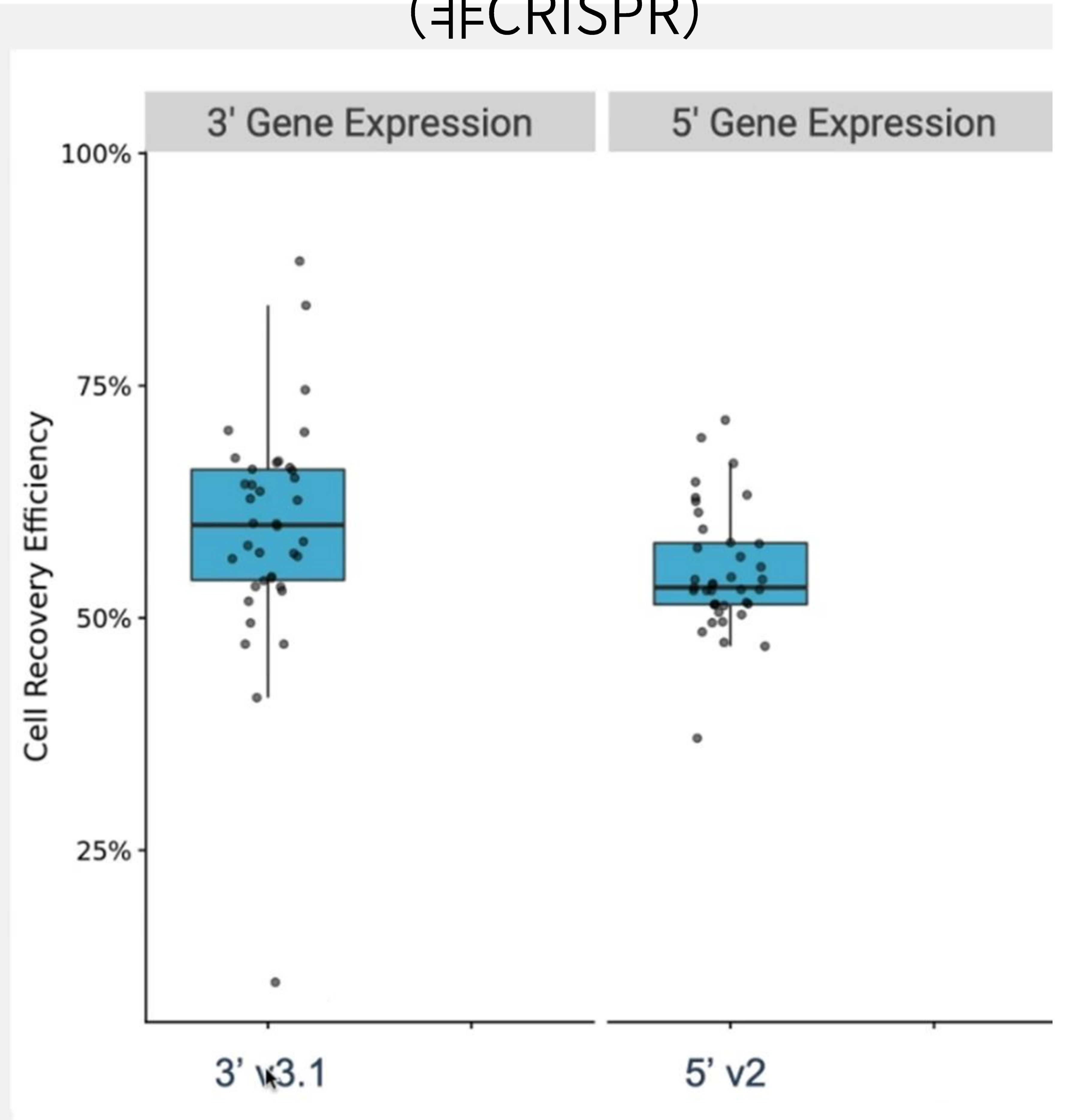
关于捕获效率

- ➤ 10×官网Demo数据统计：

序号	名称	sgRNA文库	细胞	上机细胞数	捕获细胞数	1种及以上sgRNA细占比	2种及以上sgRNA细占比	1种sgRNA细占比
1	10k K562 Transduced with Small Guide Library, 5' HT v2.0, CRISPR Screening, Chromium X	18条（8条sgRNA+10条non-targeting sgRNA）	JK562 dCas9 (Organoid)	16000	7632	90.68%	37.83%	52.85%
2	10k 1:1 Mixture of Jurkat and MM1S Cells, 5' v2.0, CRISPR Screening, Chromium X	2条（non-targeting sgRNA+Rab1a sgRNA）	Jurkat and MM1 dCas9 parental cell（4种细胞1：1：1：1混合）	16000	9932	77.23%	2.67%	74.56%
3	20k 1:1 Mixture of Jurkat and MM1S Cells, 5' HT v2.0, CRISPR Screening	2条（non-targeting sgRNA+Rab1a sgRNA）	Jurkat and MM1 dCas9 parental cell（4种细胞1：1：1：1混合）	30000	17964	80.54%	2.45%	78.09%
4	5k A549, Lung Carcinoma Cells, No Treatment Transduced with a CRISPR Pool	93条（90条sgRNA+3条non-targeting sgRNA）	A549 lung carcinoma cells that expressed dCas9-KRAB	8300	5867	82.96%	24.29%	58.67%

捕获效率47.7%-70.7%

单细胞转录组测序
（非CRISPR）



X仪器standard芯片

scCRISPR筛选技术优势：

1.将CRISPR编辑与细胞表型联系起来：对已经被sgRNA文库干扰过的细胞群，在检测mRNA基因表达的同时，同时针对sgRNA做检测，把单个细胞的表达谱与这个细胞中的sgRNA关联起来。

2.提高实验效率：用sgRNA文库，对细胞群做一次sgRNA干扰，就可以得到数以千计的细胞的sgRNA的干扰后的表达谱，极大地提高了基因干扰实验的实验效率。

3.实验可控：可以靶向任何CRISPR能够编辑的基因，并且可以仅根据转录组信息了解哪些细胞获得了治疗。

4.实验周期短：因为读数是转录表型，只要CRISPR扰动产生了基因表达层面的变化，就可以对细胞进行实验，无需长期培养以达到肉眼可见的细胞状态的变化，大大缩短了细胞培养的时间。

5.减少批次效应：实验组（转染了guide RNA的细胞）和对照组（未转染guideRNA的细胞）可以在同一实验中进行制备。

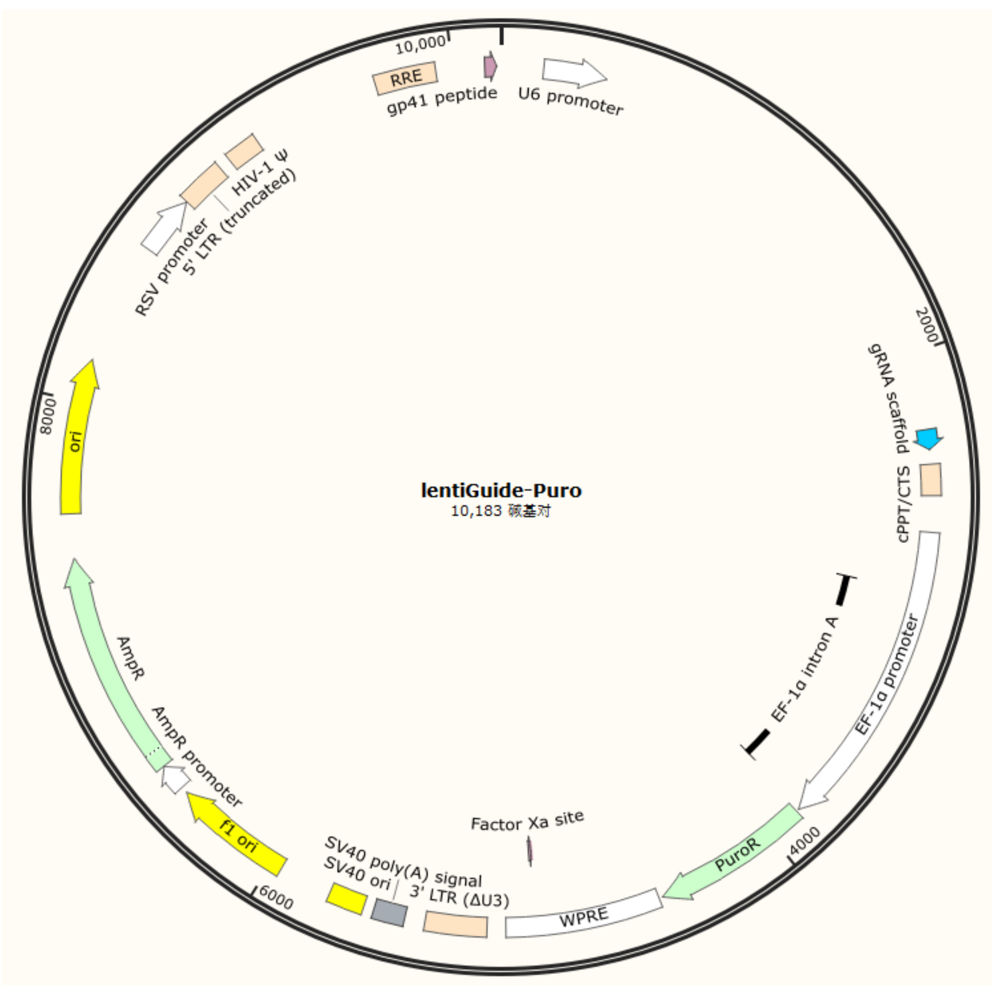
6.简化分析流程：可同时分析多个靶标以及多个样本，并将筛选的多个靶标信息对应到单个细胞上。

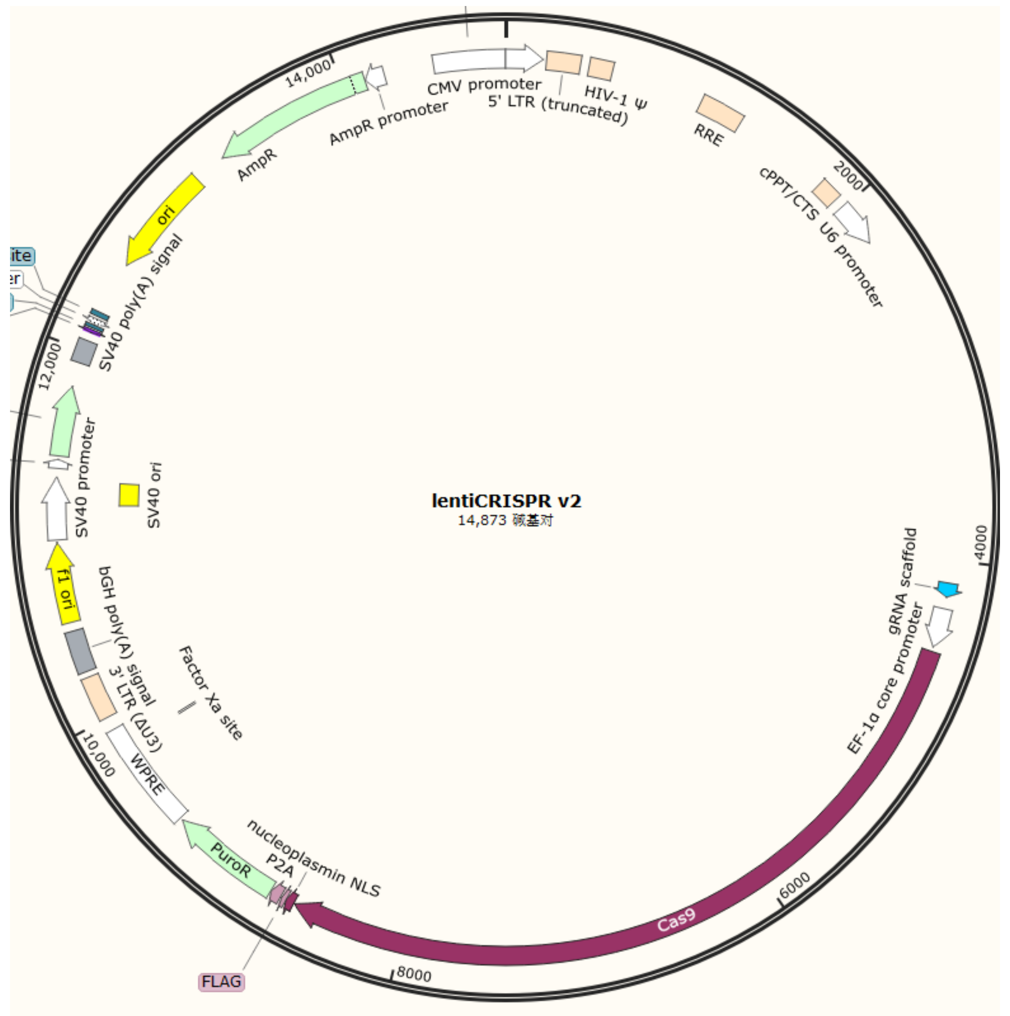
构建sgRNA质粒文库

优势：

- Oligo Pool合成——高保真性高均一性
- 可提供Broad研究所专利授权的CRISPR-Cas9文库载（LentiCRISPR v2和LentiGuide-Puro），也可以克隆至客户指定的载体中。
- 具有丰富的文库构建技术经验和成熟的NGS测序平台，确保文库质量。
- 严格的质检标准：

  覆盖度：≥99%；均一性：＜8。





✖ 不推荐

案例展示（100条sgRNA文库）

Table 4.2.1 样本比对信息统计表

Sample	sg100
Read	9418005
Mappedreads	9125272(96.89%)
Grna_mean_depth	91252
Max_depth	176156
Median_depth	89893
Totalgrnas	100 → 无sgRNA丢失
Zerogrnas	0
Coverage_rate	100.00% → sgRNA覆盖率达到100%
Totalgenes	100
Zerogenes	0
Giniindex	0.01504

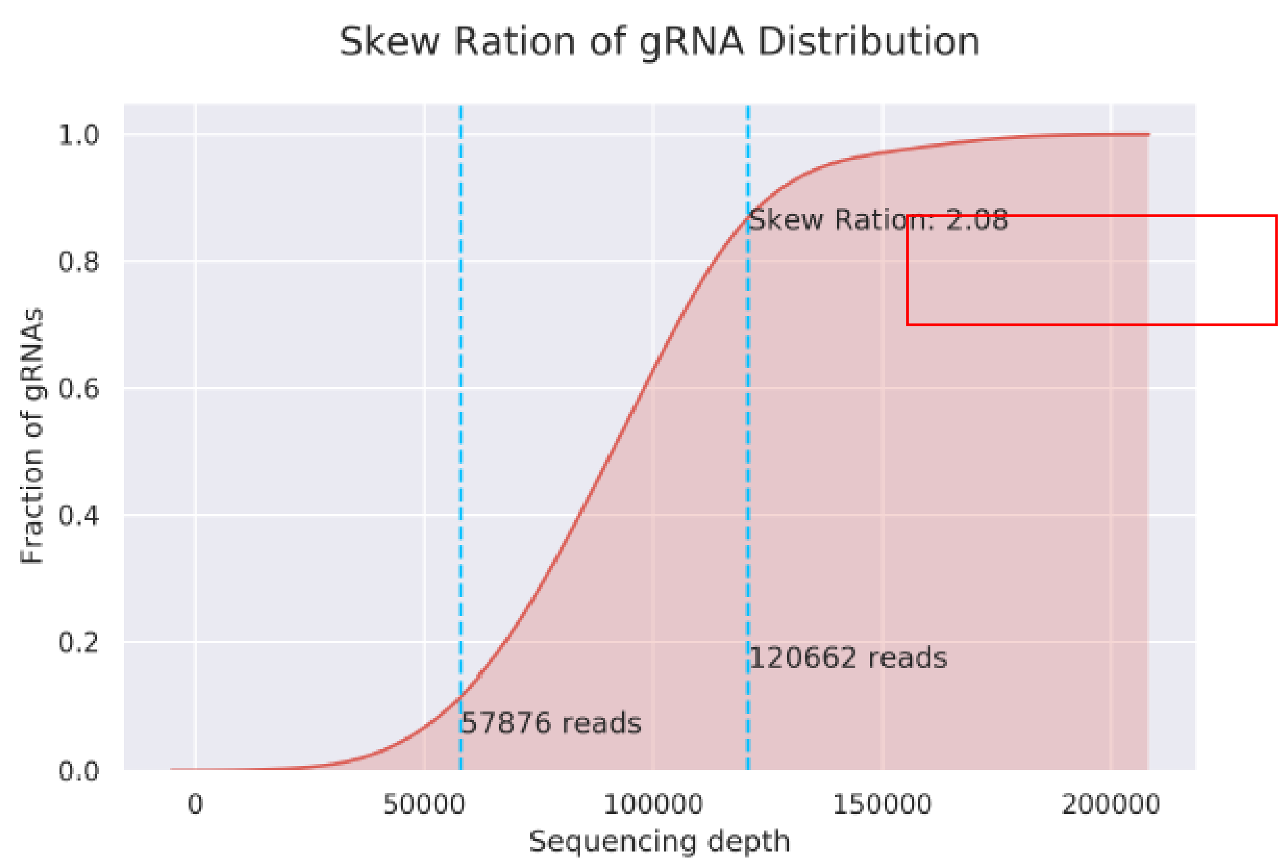


Figure 4.3.2 均一性斜率图

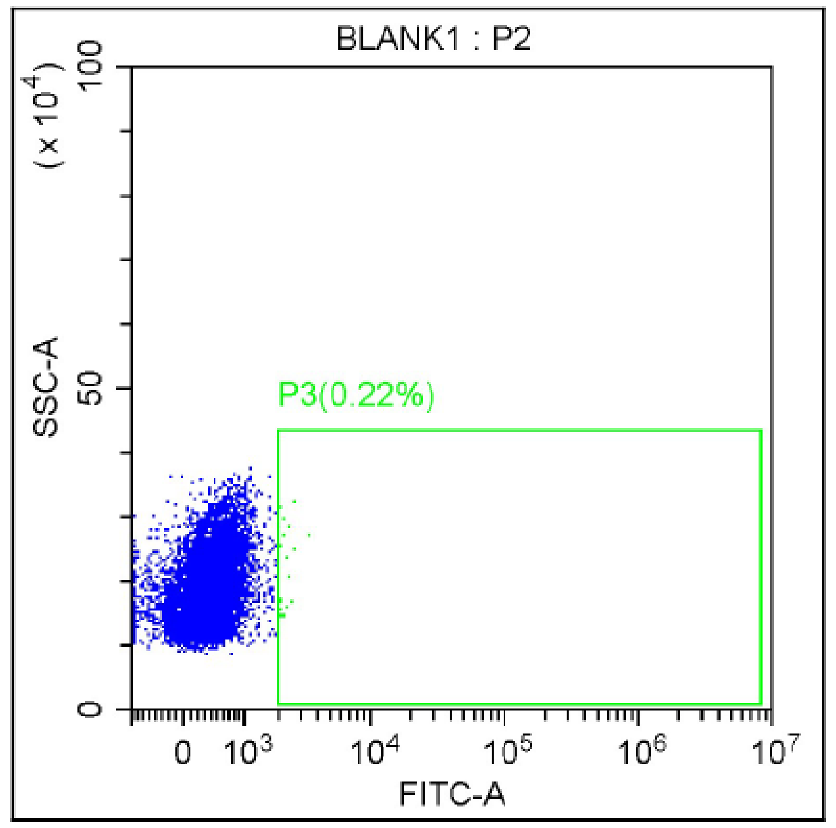
慢病毒包装、感染及细胞筛选

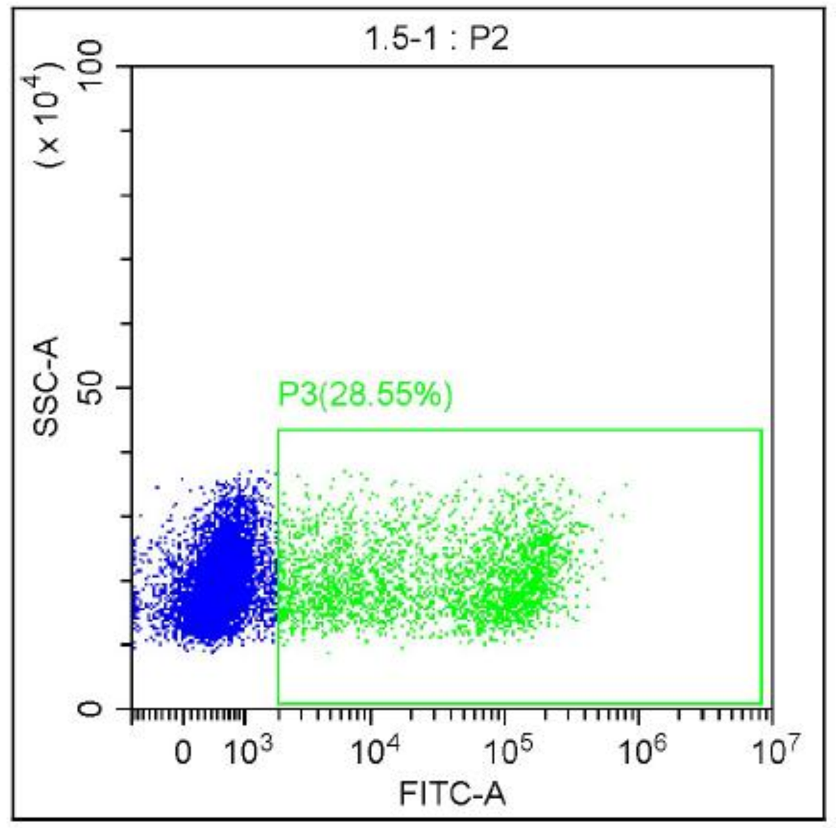
案例展示（100条sgRNA文库）

➢慢病毒感染48h后，分别进行流式分析及加入puro抗性进行筛选。

滴度测定	稀释计数	梯度稀释病毒，依次感染细胞，在荧光显微镜下观察细胞荧光占比，获得大致滴度数值
	流式分析	感染后48小时，荧光充分表达，不加药筛，被感染和未感染细胞都生长，通过流式分析荧光占比，得出具体滴度数值

低MOI感染（<0.3）



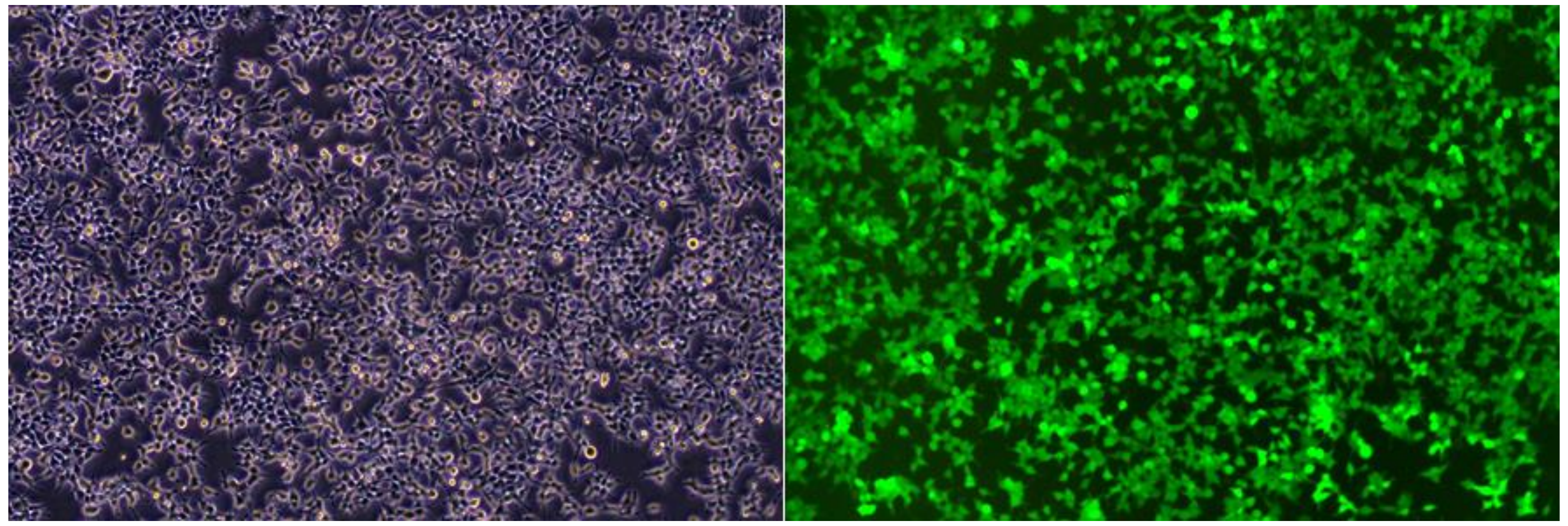


上机前细胞 QC

多个细胞转染的sgRNA多样性和单一细胞sgRNA种类数检测：

1.分单细胞培养后，多个单克隆细胞分别提取核酸，检测单克隆sgRNA序列是否为一条和多个单克隆sgRNA种类数多样性。
缺点：培养实验周期长，检测周期2周以上。
2. 用流式或稀释法，将单个细胞分到96孔板中，做微量细胞RT扩增，检测每个细胞中有几种sgRNA及多个细胞中sgRNA种类数。
优点：检验周期快，最快36h即可告知结果，判断细胞转染符合要求后即可安排10x上机流程。

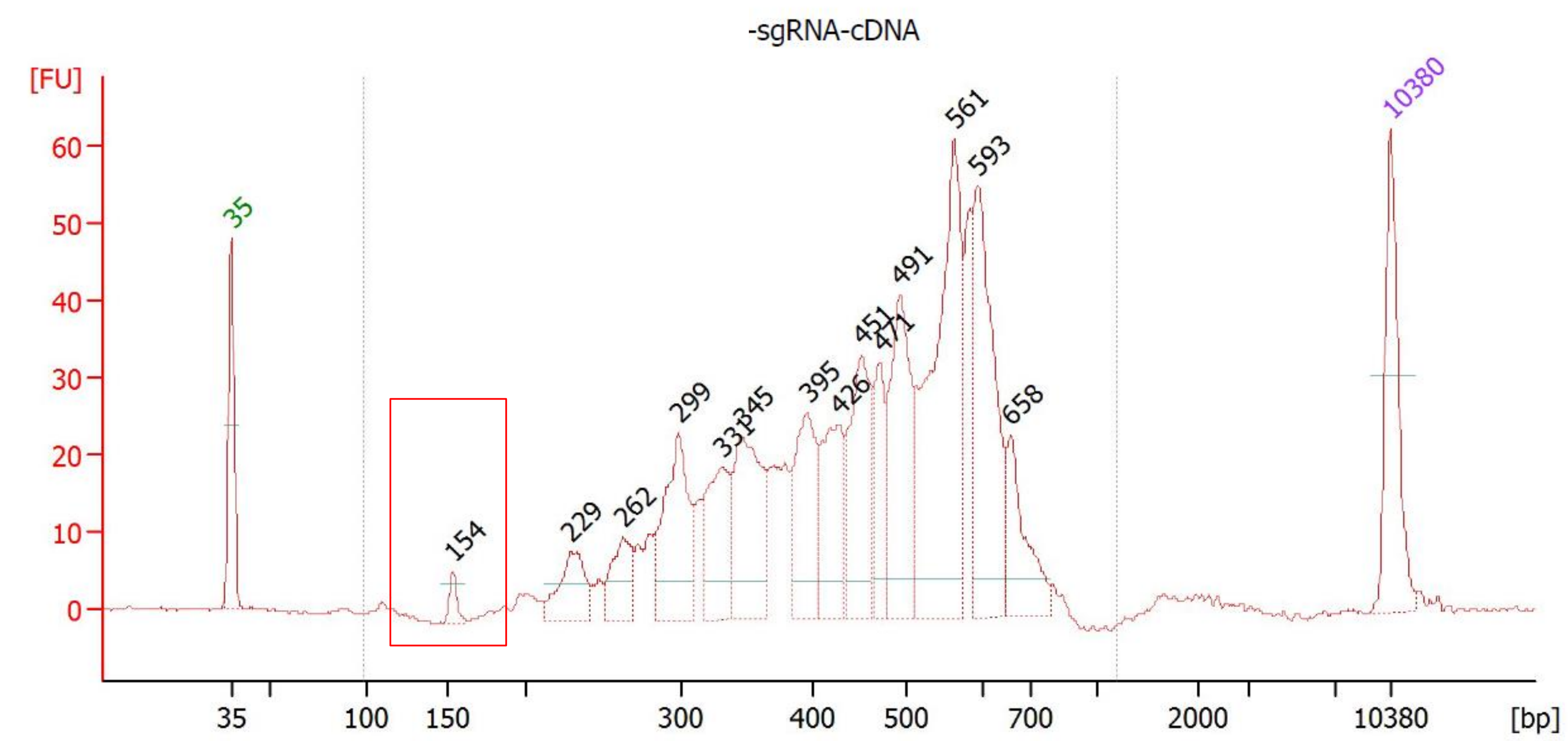
➢puro筛选5天，准备上机。

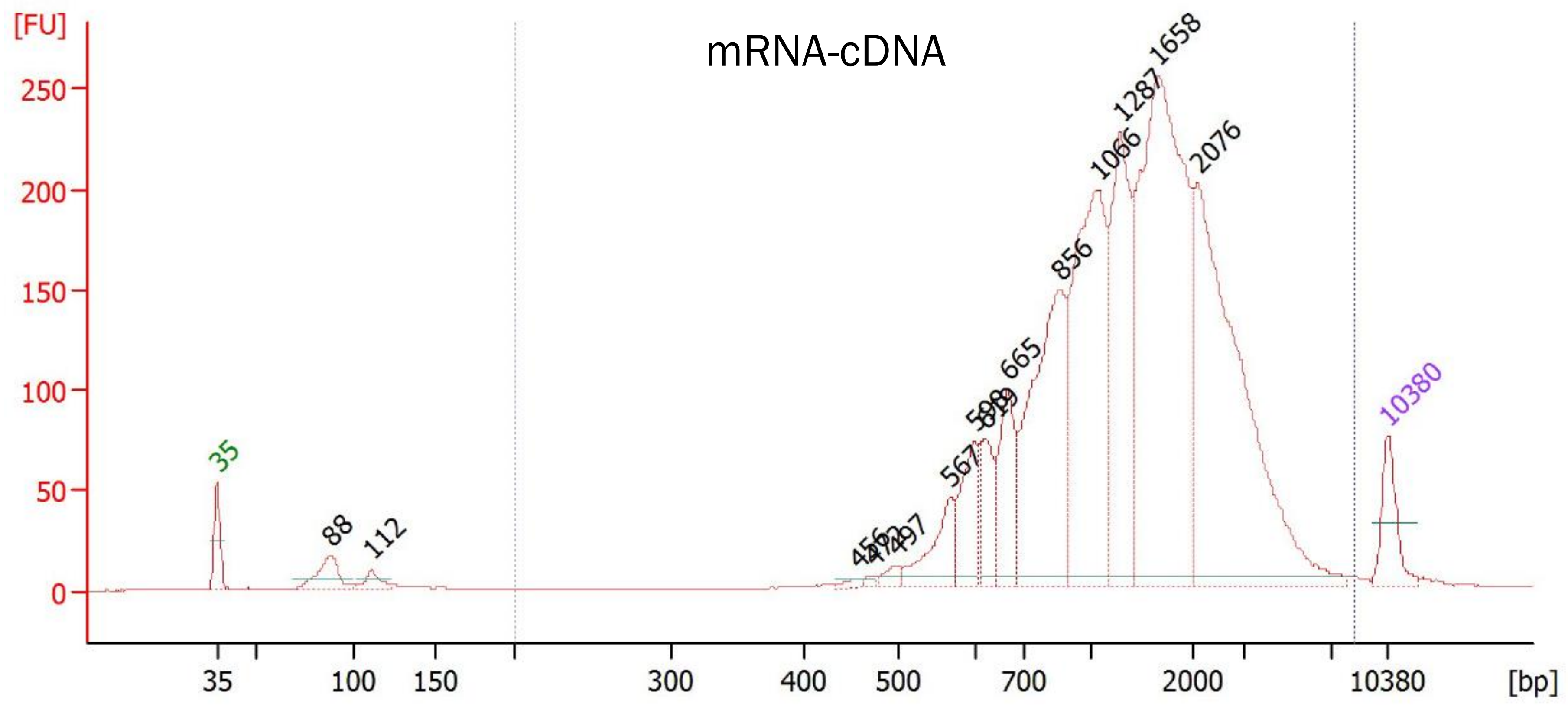


➢上机前细胞状态检测。

上机前细胞状态：
- 细胞活率：97.91%
- 结团率6%

cDNA和文库QC


-sgRNA-cDNA
[FU]
[bp]
35
154
229
262
299
561
593
658
10380


mRNA-cDNA
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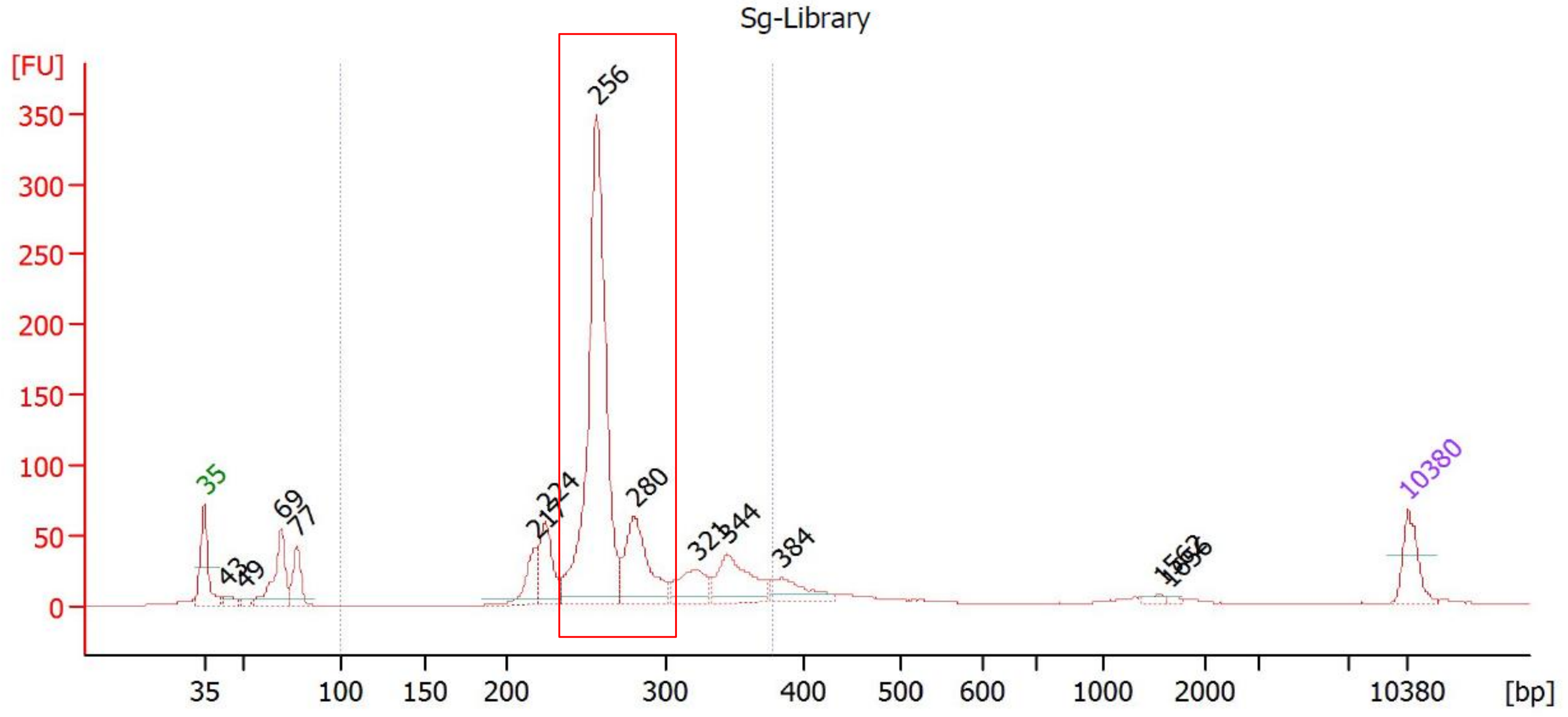
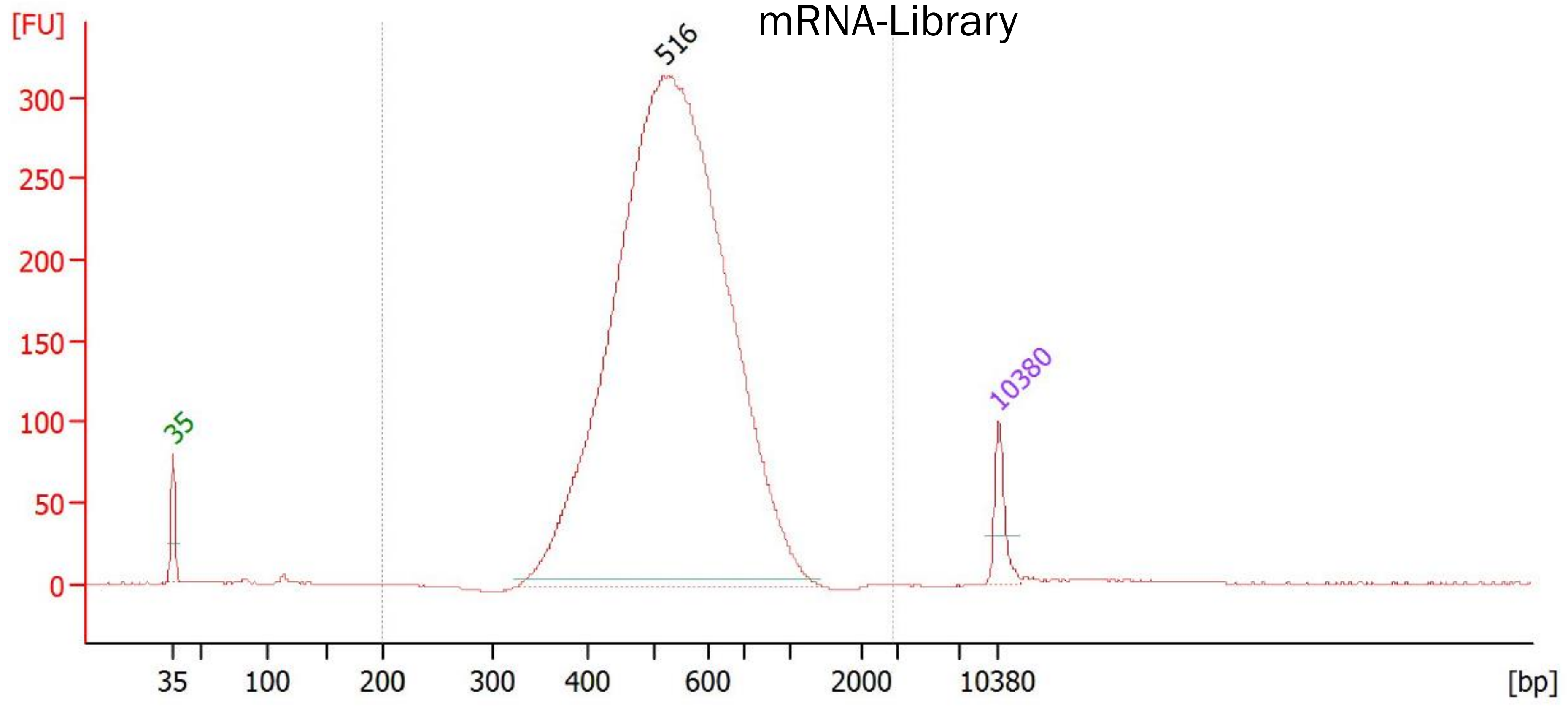
112
567
665
856
1066
1287
1658
2076
10380


Sg-Library
[FU]
[bp]
35
256
280
384
10380


mRNA-Library
[FU]
[bp]
35
516
10380

案例展示1

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Summary | Analysis

11,730
Estimated Number of Cells

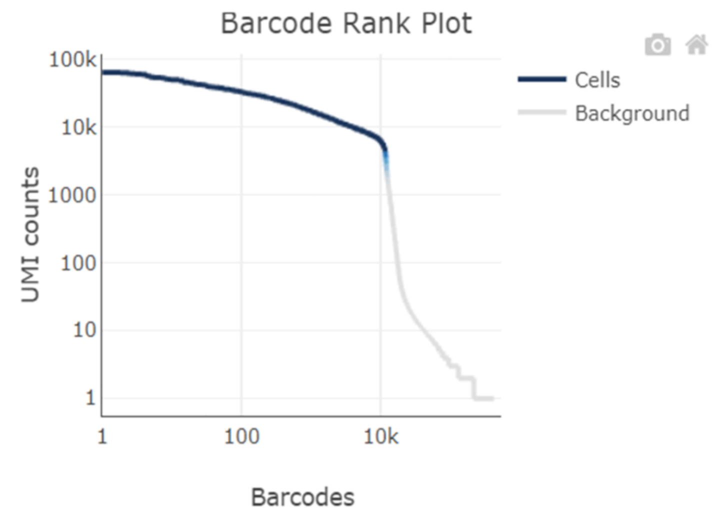
26,292
Mean Reads per Cell

2,878
Median Genes per Cell

Sequencing

Number of Reads	308,401,314
Number of Short Reads Skipped	0
Valid Barcodes	92.9%
Valid UMIs	100.0%
Sequencing Saturation	47.4%
Q30 Bases in Barcode	96.4%
Q30 Bases in RNA Read	72.5%
Q30 Bases in RNA Read 2	89.8%
Q30 Bases in UMI	96.4%

Cells



Estimated Number of Cells	11,730
Fraction Reads in Cells	95.5%
Mean Reads per Cell	26,292
Median Genes per Cell	2,878
Total Genes Detected	22,953
Median UMI Counts per Cell	8,382

CRISPR Sequencing

Number of Reads	36,164,901
Number of Short Reads Skipped	0
Valid Barcodes	97.3%
Valid UMIs	100.0%
Sequencing Saturation	90.9%
Q30 Bases in Barcode	96.8%
Q30 Bases in CRISPR Read	69.1%
Q30 Bases in CRISPR Read 2	94.3%
Q30 Bases in UMI	96.6%

CRISPR Application

Fraction Reads with Putative Protospacer Sequence	92.6%
Fraction Guide Reads	91.2%
Fraction Guide Reads Usable	83.2%
Guide Reads Usable per Cell	2,566
Fraction Protospacer Not Recognized	1.5%
Guide Reads in Cells	93.4%
Cells with 1 or more protospacers detected	94.8%
Cells with 2 or more protospacers detected	24.5%
Median UMIs per Cell	137

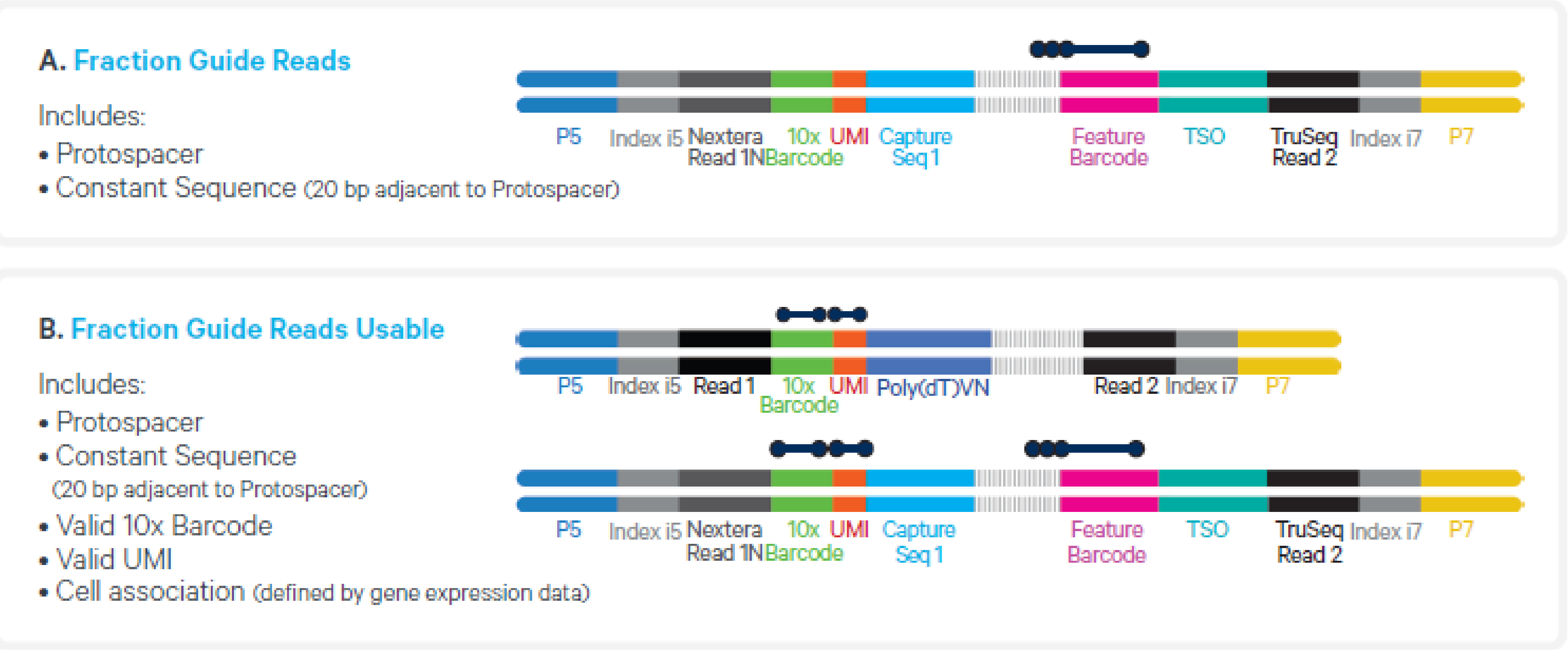


Figure 12. Schematic showing Fraction Guide Read (A) and Fraction Guide Reads Usable (B) parameters for guide RNA based calling.

CC仪器

- 下机数据：mRNA文库85G，sgRNA文库10.2G。
- 上样量1.8W，捕获率约66.7%；
- 含sgRNA细胞数目占比94.8%；
- 单种sgRNA细胞数目占比70.3%。

案例展示2

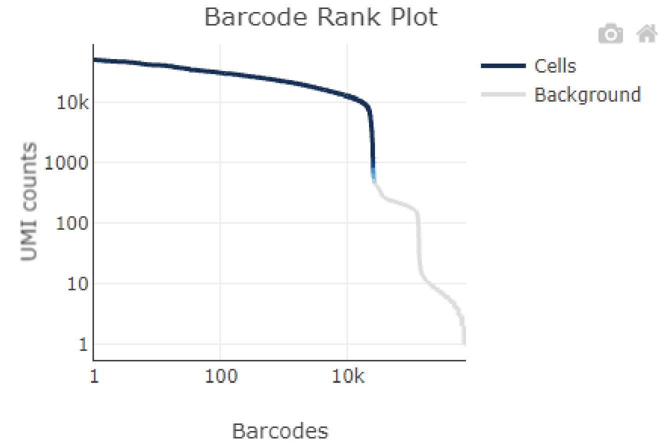
Summary Analysis

25,764	Estimated Number of Cells
26,120	Mean Reads per Cell
3,687	Median Genes per Cell

Sequencing

Number of Reads	672,963,484
Number of Short Reads Skipped	0
Valid Barcodes	90.5%
Valid UMIs	99.8%
Sequencing Saturation	18.4%
Q30 Bases in Barcode	95.0%
Q30 Bases in RNA Read	83.8%
Q30 Bases in RNA Read 2	89.0%
Q30 Bases in UMI	94.9%

Cells



Estimated Number of Cells	25,764
Fraction Reads in Cells	92.0%
Mean Reads per Cell	26,120
Median Genes per Cell	3,687
Total Genes Detected	19,275
Median UMI Counts per Cell	11,552

CRISPR Sequencing

Number of Reads	63,846,651
Number of Short Reads Skipped	0
Valid Barcodes	97.3%
Valid UMIs	100.0%
Sequencing Saturation	62.3%
Q30 Bases in Barcode	96.3%
Q30 Bases in CRISPR Read	89.6%
Q30 Bases in CRISPR Read 2	94.9%
Q30 Bases in UMI	95.7%

CRISPR Application

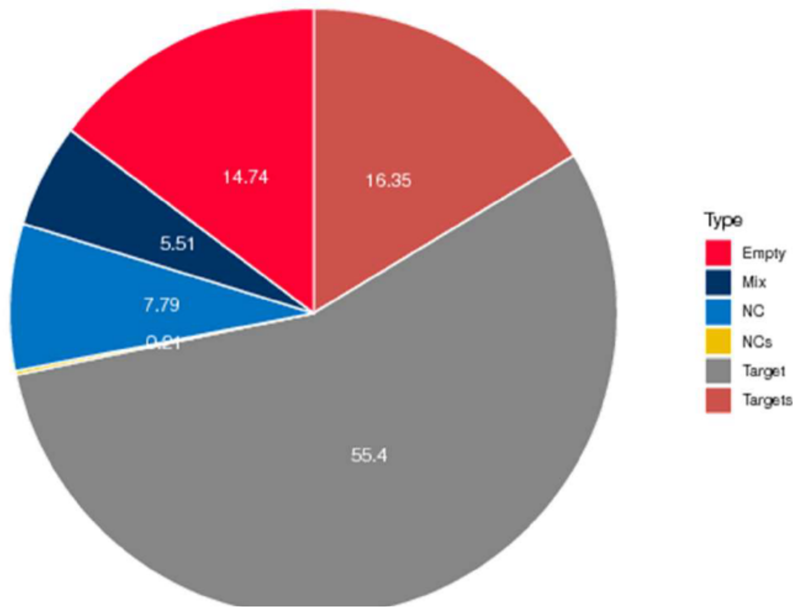
Fraction Reads with Putative Protospacer Sequence	88.6%
Fraction Guide Reads	85.0%
Fraction Guide Reads Usable	69.2%
Guide Reads Usable per Cell	1,714
Fraction Protospacer Not Recognized	4.1%
Guide Reads in Cells	83.2%
Cells with 1 or more protospacers detected	86.5%
Cells with 2 or more protospacers detected	23.1%
Median UMIs per Cell	270

X仪器

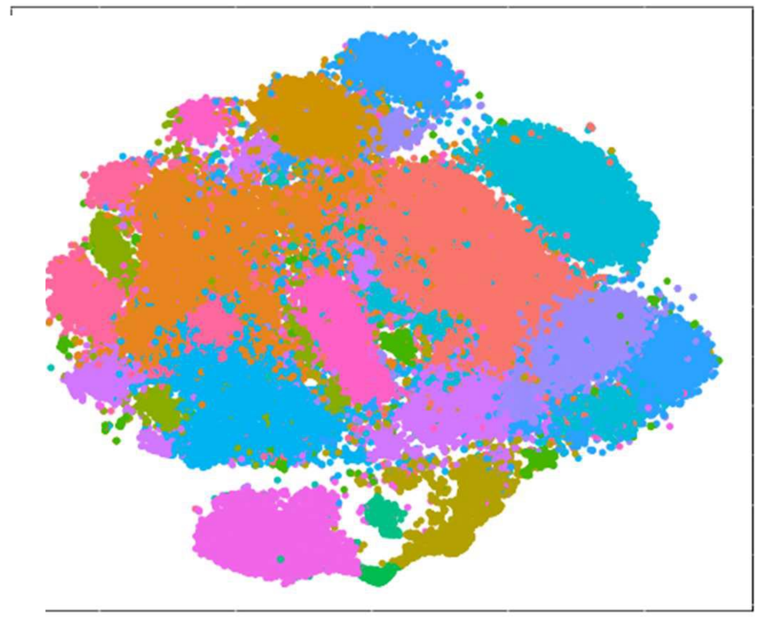
- 下机数据：mRNA文库201.9G，sgRNA文库19.2G。
- 上样量3.8W，捕获率约67.8%；
- 含sgRNA细胞数目占比86.5%；
- 单种sgRNA细胞数目占比63.4%。

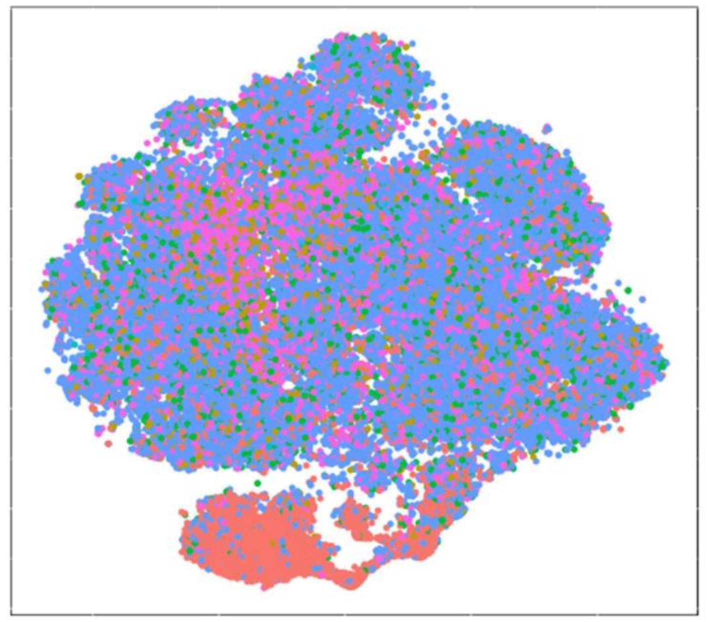
表 4. 细胞内 gRNA 分布表

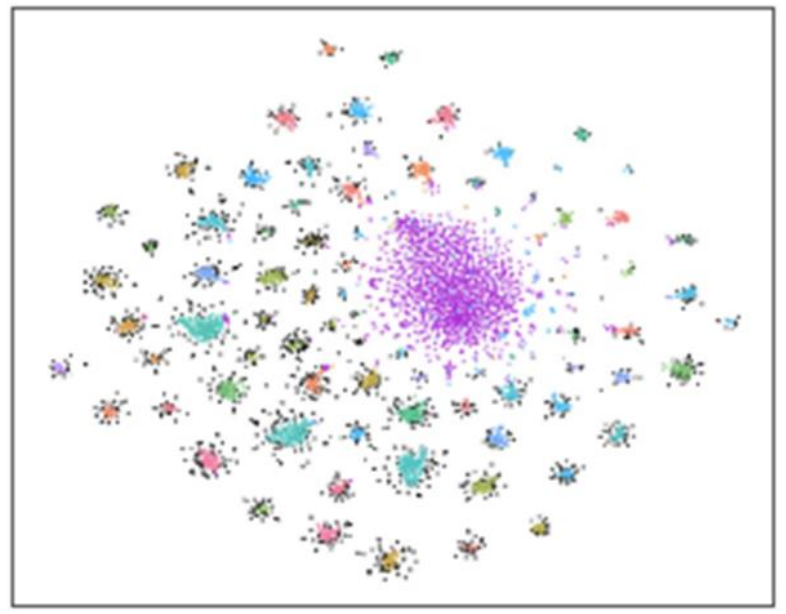
Cells	gRNA	TargetGene	gRNAsInCell	GenesInCell
AAACCTGAGACTTTCG-1	gRNA48;	Nkx3-1;	1	2699
AAACCTGAGAGACGAA-1	gRNA73;	Agmat;	1	4498
AAACCTGAGATATGGT-1	gRNA11;gRNA100;	D1Ertd622e;NCsgRNA-1;	2	4730



细胞内 gRNA 分布比例图


t-SNE 聚类


gRNA 在细胞分群中的分布(t-SNE)

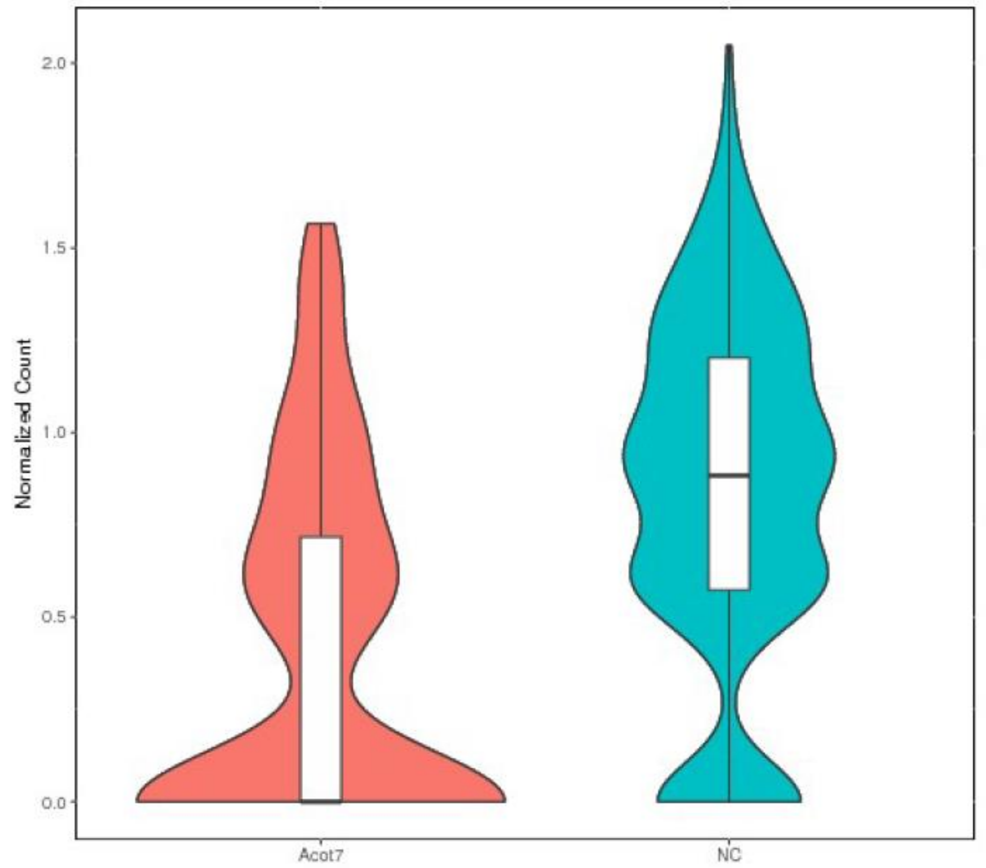

CRISPR 聚类

细胞中所有基因的表达矩阵（示例）

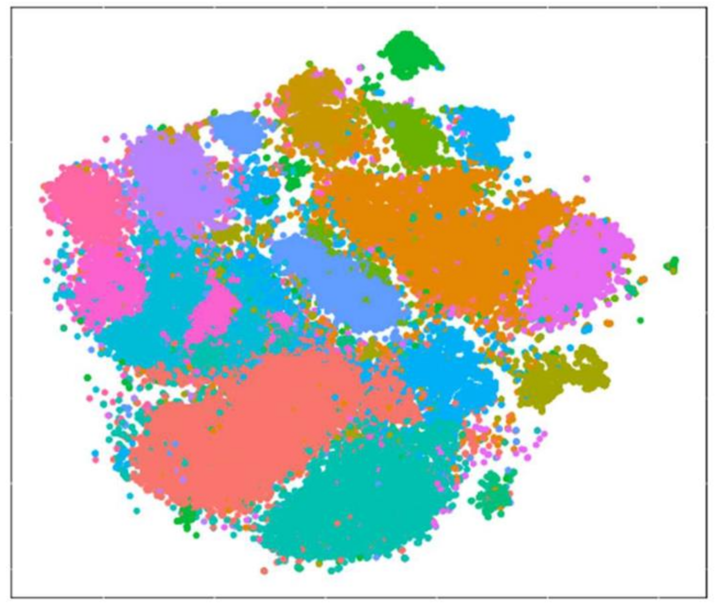
Gene	AAACCTGAGACTTTCG-1
Xkr4	0
Gm1992	0
Gm19938	0
Gm37381	0
Rp1	0
Sox17	0
Gm37587	0
Gm37323	0
Mrpl15	3
Lypla1	0

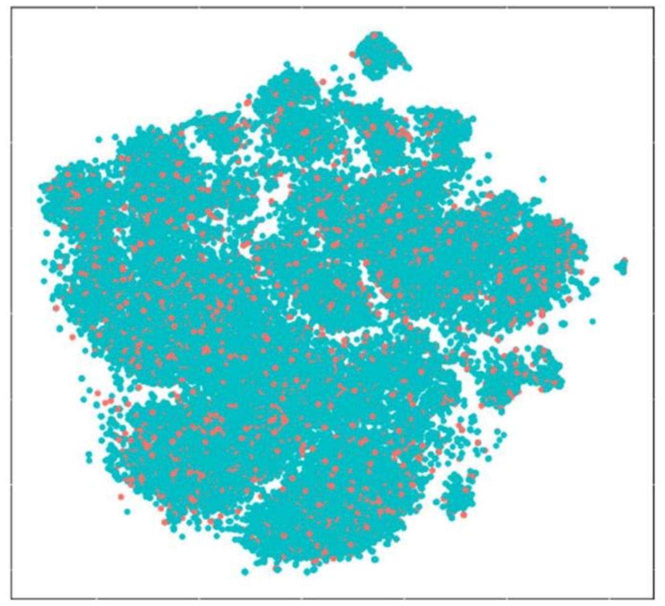
表 7. 每类细胞群 gRNA 丰度表（示例）

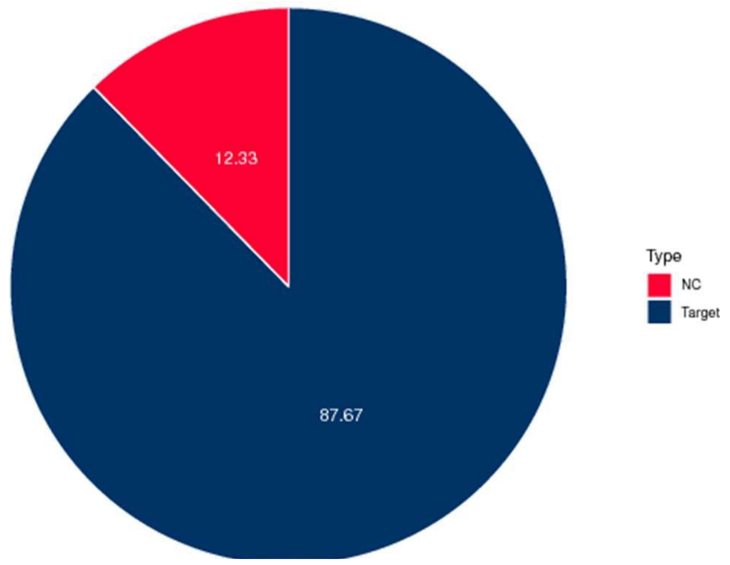
ID	Group0	Group1	Group10	Group11	Group12	Group13	Group14	Group15	Group16
gRNA1	154	172	46	11	18	7	2	5	76
gRNA10	143	146	40	23	15	23	1	2	4
gRNA100	188	183	54	28	19	19	2	2	14



不同分类的细胞中 Target 基因表达量（以第一类细胞 Acot7 为例）

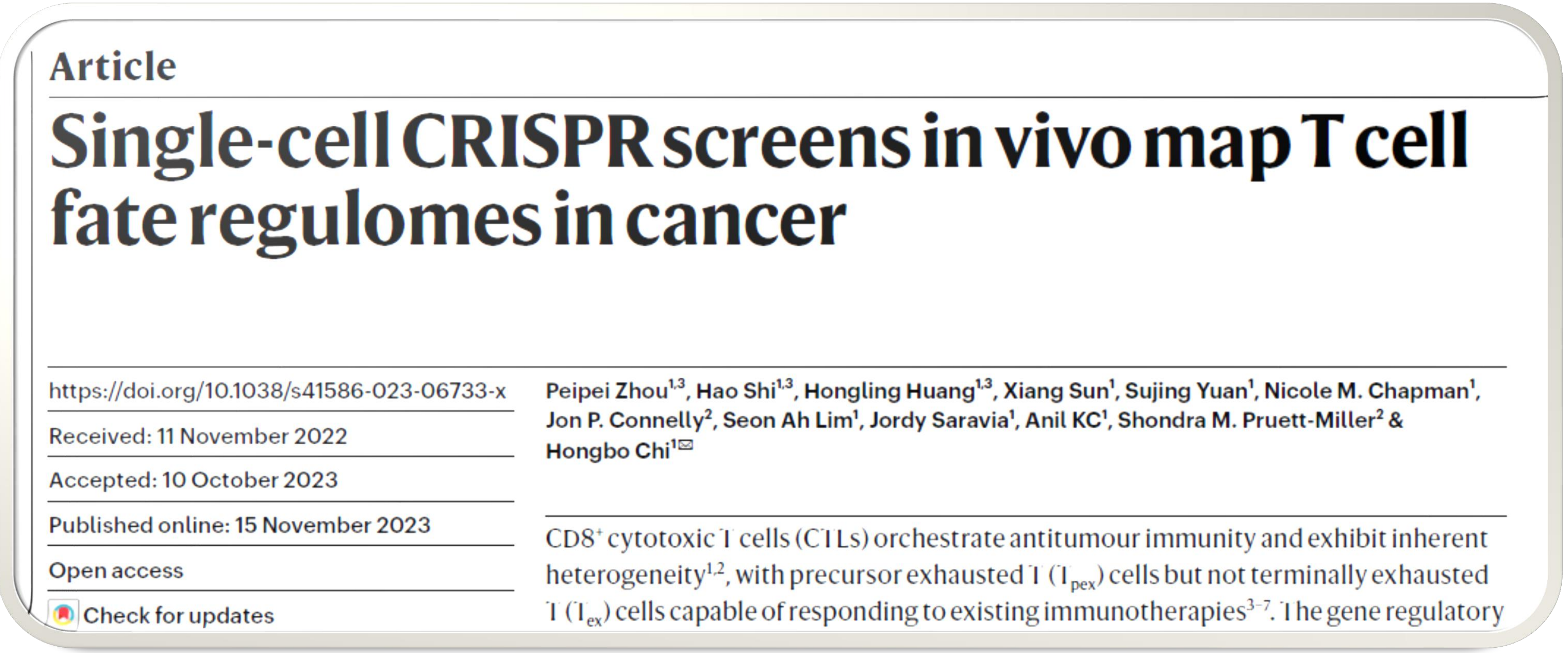

t-SNE 聚类(过滤后)


gRNA 在细胞分群中的分布(过滤后 t-SNE)



细胞内 gRNA 分布比例图（过滤后）

scCRISPR应用


Article

Single-cell CRISPR screens in vivo map T cell fate regulomes in cancer

https://doi.org/10.1038/s41586-023-06733-x

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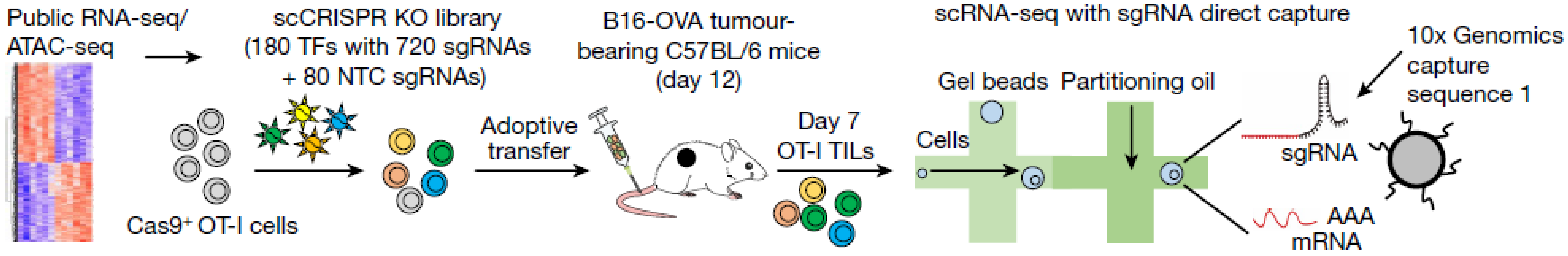
Check for updates

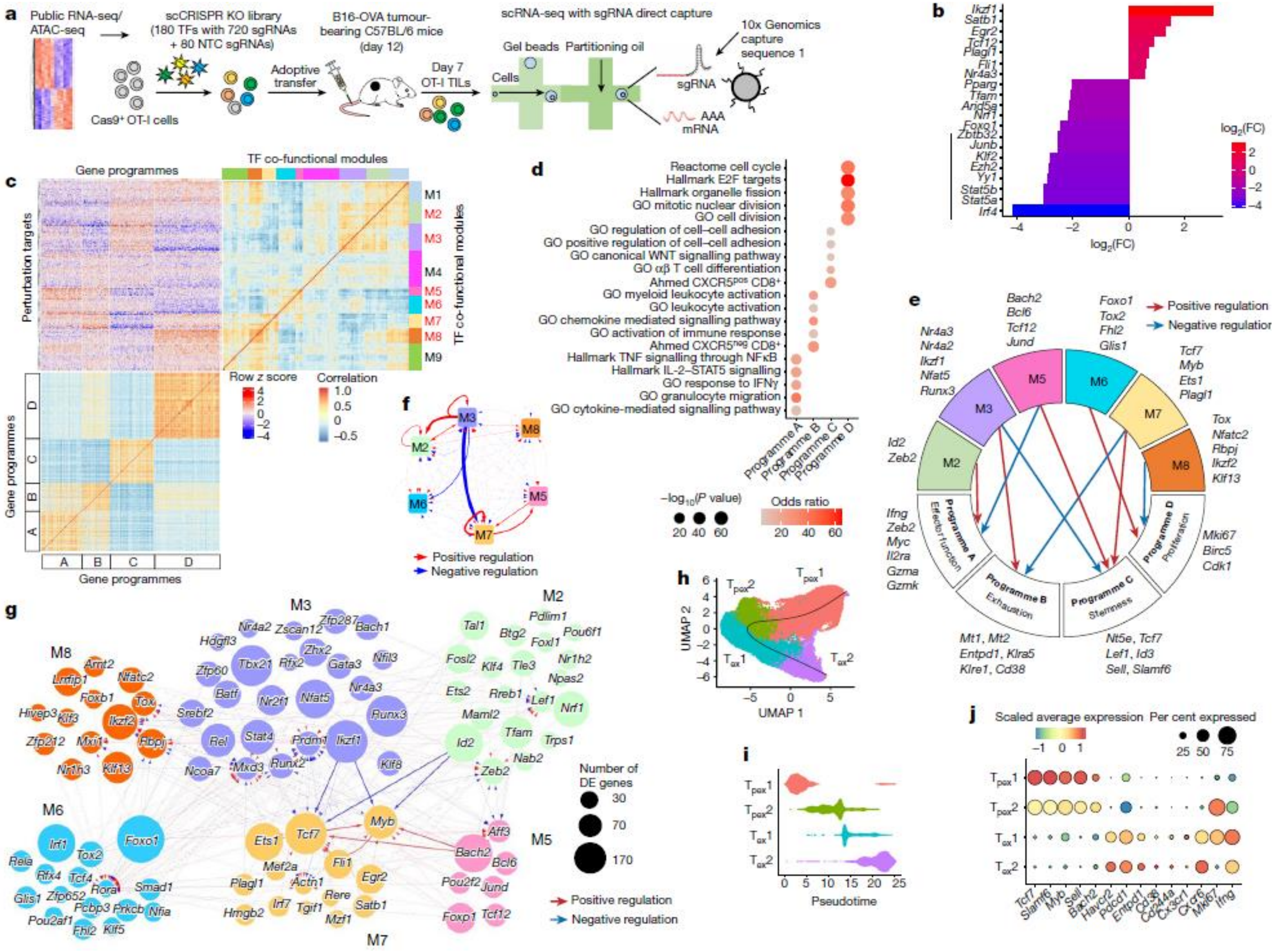
Peipei Zhou[1,3], Hao Shi[1,3], Hongling Huang[1,3], Xiang Sun[1], Sujing Yuan[1], Nicole M. Chapman[1], Jon P. Connelly[2], Seon Ah Lim[1], Jordy Saravia[1], Anil KC[1], Shondra M. Pruett-Miller[2] & Hongbo Chi[1]✉

CD8[+] cytotoxic T cells (CTLs) orchestrate antitumour immunity and exhibit inherent heterogeneity[1,2], with precursor exhausted T (T_{pex}) cells but not terminally exhausted T (T_{ex}) cells capable of responding to existing immunotherapies[3–7]. The gene regulatory



Zhou, P., Shi, H., Huang, H. *et al.* Single-cell CRISPR screens in vivo map T cell fate regulomes in cancer. *Nature* (2023). https://doi.org/10.1038/s41586-023-06733-x

scCRISPR应用



scCRISPR应用

关于捕获细胞数目

Sequencing library preparation. Sorted OT-I cells were resuspended and diluted in 1× PBS (Thermo Fisher Scientific) containing 0.04% BSA (Amresco) at a concentration of 1×10^6 cells per ml. Both the gene expression library and the CRISPR screening library were prepared using a Chromium Next GEM Single Cell 3′ kit with Feature Barcode technology for CRISPR Screening (v.3.1; 10x Genomics). In brief, the single-cell suspensions were loaded onto the Chromium Controller according to their respective cell counts to generate 10,000 single-cell gel beads in emulsion per sample. Each sample was loaded into four separate channels. The resulting libraries were quantified and quality checked using TapeStation (Agilent). Samples were diluted and loaded onto a NovaSeq (Illumina) to a sequencing depth of 500 million reads per channel for gene expression libraries and 200 million reads per channel for CRISPR screening libraries.

age (>10%) of mitochondrial reads were also removed. For the sg*Ikzf1* experiment, a total of 28,945 cells (sgNTC-transduced, 14,044 cells; sg*Ikzf1*-transduced, 14,901 cells) were captured, with an average of 3,012 mRNA molecules (UMIs, median: 11,906; range: 1,026–49,991). For the sg*Ets1* experiment, a total of 20,618 cells (sgNTC-transduced, 10,562 cells; sg*Ets1*-transduced, 10,056 cells) were captured, with an average of 2,955 mRNA molecules (UMIs, median: 12,477; range: 1,703–49,993). For the sg*Rbpj* experiment, a total of 19,516 cells (sgNTC-transduced, 10,918 cells; sg*Rbpj*-transduced, 8,598 cells) were captured, with an average of 2,908 mRNA molecules (UMIs, median: 11,327; range: 1,265–49,887). The expression data were normalized using the NormalizeData function in Seurat with scale.factor = 10^6. Raw and processed scRNA-seq data have been deposited into the GEO database with the series identifier GSE216800.

3’ CRISPR试剂盒，CC仪器，每通道目的捕获10000个细胞，每个样本4个通道，实际捕获19516-28945个细胞。即CC仪器每次捕获4879-7237个细胞。

Zhou, P., Shi, H., Huang, H. *et al.* Single-cell CRISPR screens in vivo map T cell fate regulomes in cancer. *Nature* (2023). https://doi.org/10.1038/s41586-023-06733-x

scCRISPR应用

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> Nat Neurosci. 2021 Jul;24(7):1020-1034. doi: 10.1038/s41593-021-00862-0. Epub 2021 May 24.

Genome-wide CRISPRi/a screens in human neurons link lysosomal failure to ferroptosis

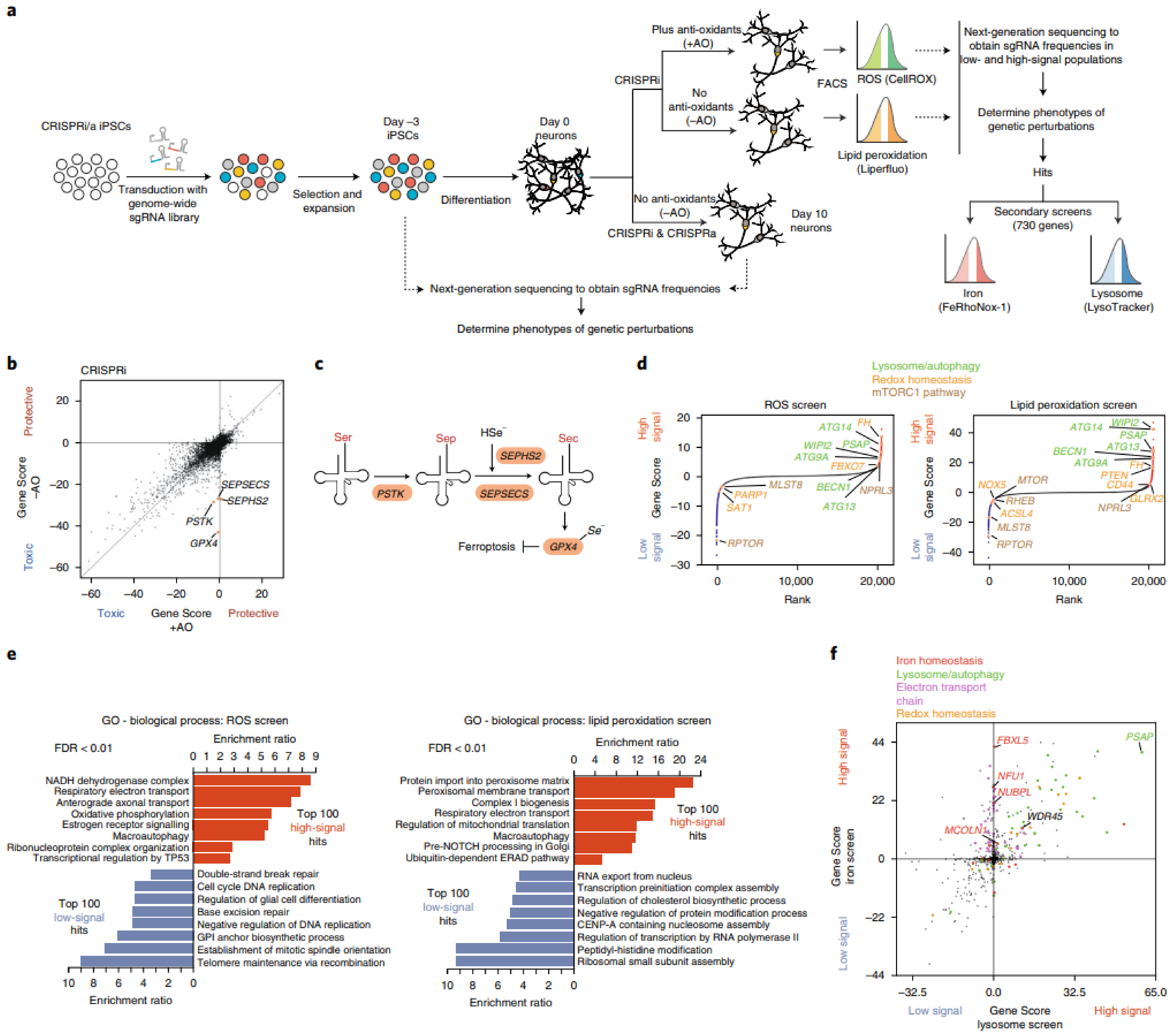
Ruilin Tian[1,2,3], Anthony Abarientos[4], Jason Hong[4], Sayed Hadi Hashemi[5], Rui Yan[6], Nina Dräger[4], Kun Leng[4], Mike A Nalls[7,8], Andrew B Singleton[7], Ke Xu[6], Faraz Faghri[5,7,8], Martin Kampmann[9,10]

Affiliations + expand

PMID: 34031600 PMCID: PMC8254803 DOI: 10.1038/s41593-021-00862-0

Free PMC article

来自加州大学旧金山分校的研究团队发表在《Nature Neuroscience》上的文章，研究团队利用混合CRISPR筛选和10x单细胞CRISPR筛选来评估来源于人诱导多能干细胞（iPSC）的神经元中的基因功能，这项研究是iPSC神经元首次在全基因组范围进行了功能基因组学评估，并且本次研究还建立了一个开源的云端数据共享的CRISPRbrain新数据库平台，能够对分化的人类细胞类型的功能基因组学筛选进行汇总，让更多的研究人员能够分享结果并合作构建图谱。



在ipsc衍生的神经元中进行基因组CRISPRi和CRISPRa筛选及单细胞测序，鉴定氧化应激存活和氧化还原稳态的调节因子

分析发现几个具有溶酶体功能但与自噬无关的强命中基因包含PSAP

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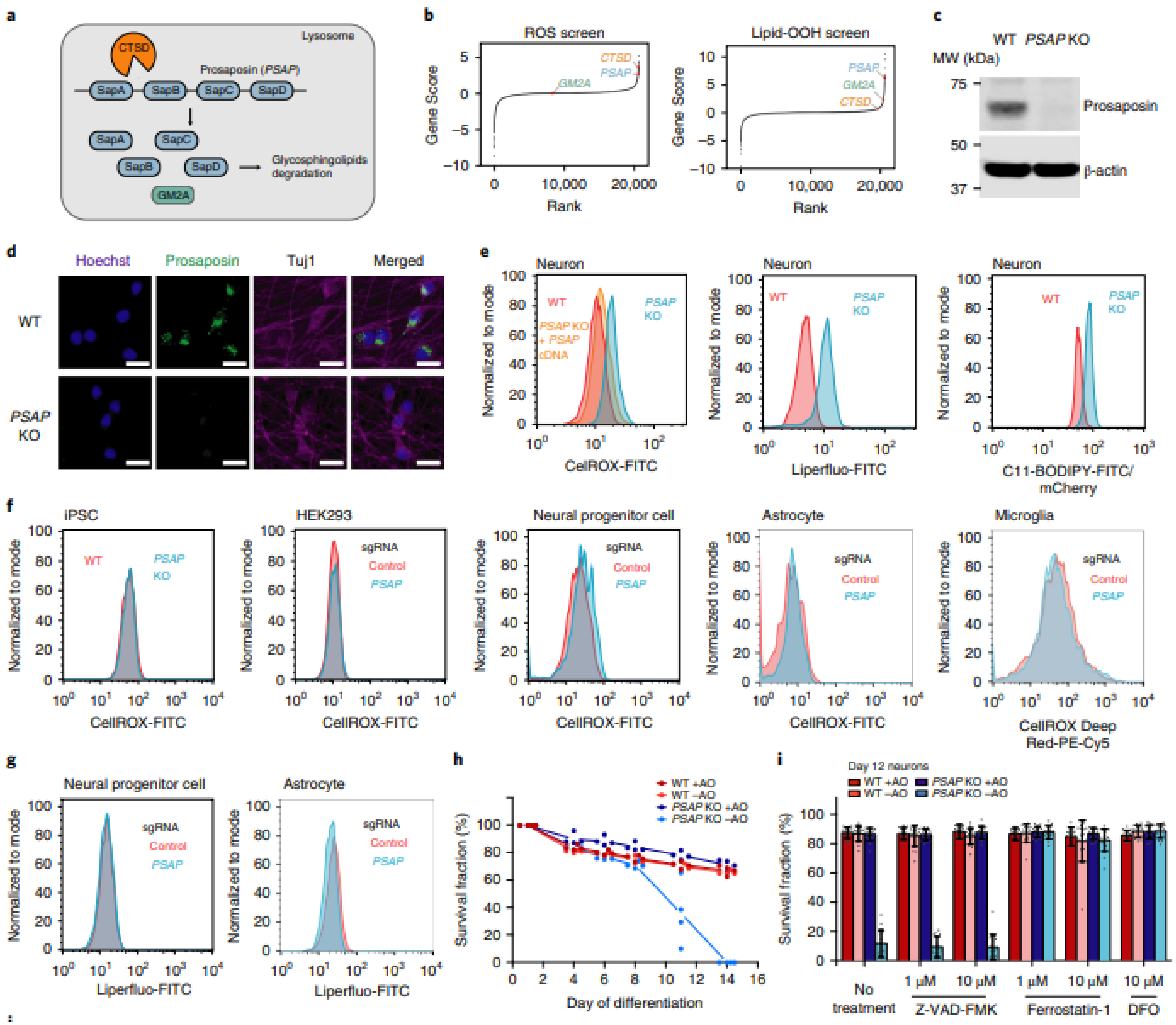
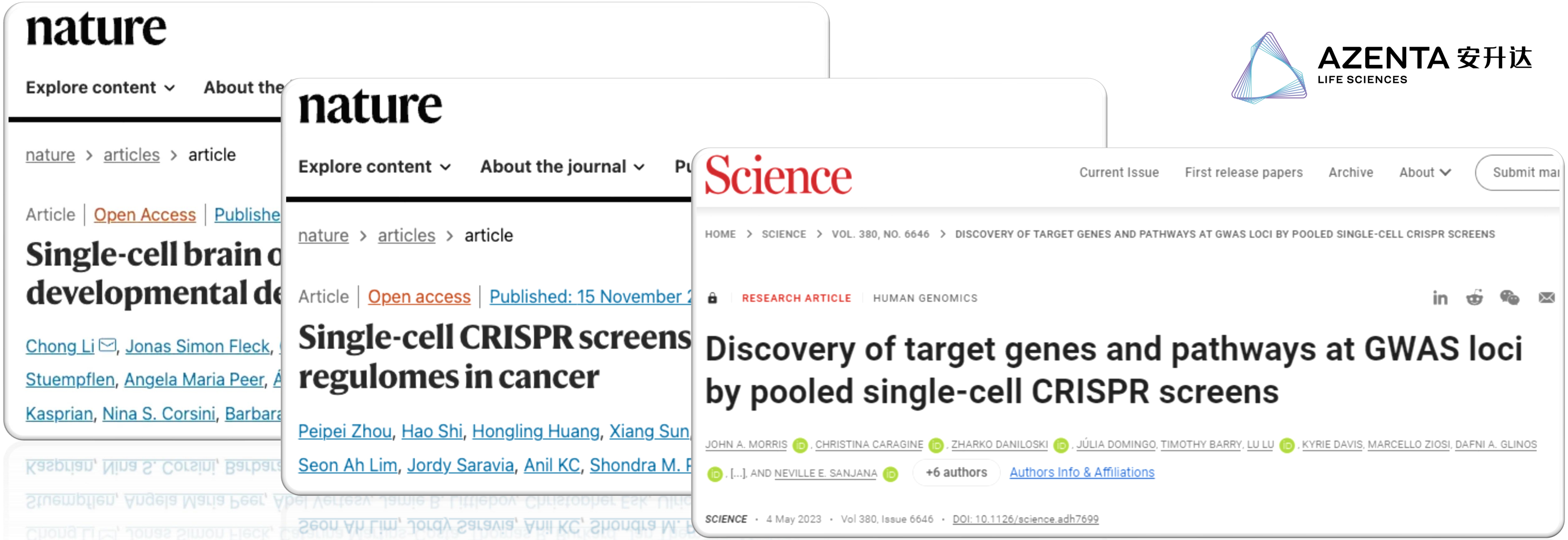


建立PSAP敲除细胞系，进行氧化应激敏感功能验证

PSAP蛋白prosaposin的缺失，神经元中的ROS和脂质会过氧化，导致神经元铁下垂

应用场景

神经发育病理学研究
病毒与宿主互作研究
基因型与表型映射
scCRISPR
探究药物作用机制
遗传调控机制研究
药物靶点开发
肿瘤生物学研究



I. 利用10x CRISPR流程，将常规细胞群CRISPR筛选与单细胞RNA测序相结合，可以更加方便地进行基因功能和基因调控网络研究，加速药物靶点的发现；

II. 10x CRISPR流程不仅可以进行常规的CRISPR编辑后的单细胞捕获，还可以利用Feature barcode捕获序列对非CRISPR基因编辑样本进行单细胞水平的靶标序列捕获。

服务优势

10x官方授权服务商

丰富的项目经验：可处理人、大小鼠及植物的多种组织样品

完善的服务流程：提供上门服务、样品冷链运输等全方位服务

专业的技术支持：专业的售前、售中及售后服务支持

一站式解决方案：针对不同研究方向提供专业的解决方案

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高通量测序 合规服务 Sanger测序 基因合成 分子生物学 引物合成 细胞和基因治疗 克隆与突变 质粒DNA制备 基因编辑
单细胞CRISPR Screen测序
单细胞CRISPR Screen可以在细胞类型或标记物未知的前提下，检测百万个细胞中影响细胞基因表达表型的sgRNA，分析多种不同的CRISPR扰动。10X单细胞测序结合Feature Barcode技术，能同时检测CRISPR扰动和由此产生的多种转录特征，可以研究靶标基因表达水平影响复杂细胞表型和进程的机制。
技术流程
01 设计 CRISPR文库
02 组装 慢病毒
03 感染并 挑选细胞
04 构建 10x Genomics文库
05 对文库 进行测序
06 获得 新见解

✓公众号


安升达生命科学
1月12日 16:34
高通量单细胞CRISPR，助力药物靶点发现新工具
药物靶点发现新工具 +1

Thank you!

AZENTA 安升达
LIFE SCIENCES